Darden Case Book

2019-2020 Edition

Darden School of Business

Consulting Club at Darden

Darden Case Book 2019-2020 Foreword

New Material + 12 Total Cases

The cases featured in the 2019-2020 Case Book are some of Darden's best cases from prior years

This case book adds a new consulting industry guide and 12 total cases

9 New Cases

The 2019-2020 Case Book features 9 brand new cases that will test a broad array of knowledge areas and math concepts

7 Writers

The 2019-2020 Darden Case Book has been made possible by Guy Schwartz, Steven McIntyre, Ryan Ritchie, Spencer Small, Aaron Wilson, Michael Hoban, and Saule Kassengaliyeva of the Darden MBA class of 2020

Darden Case Book 2019-2020 Index

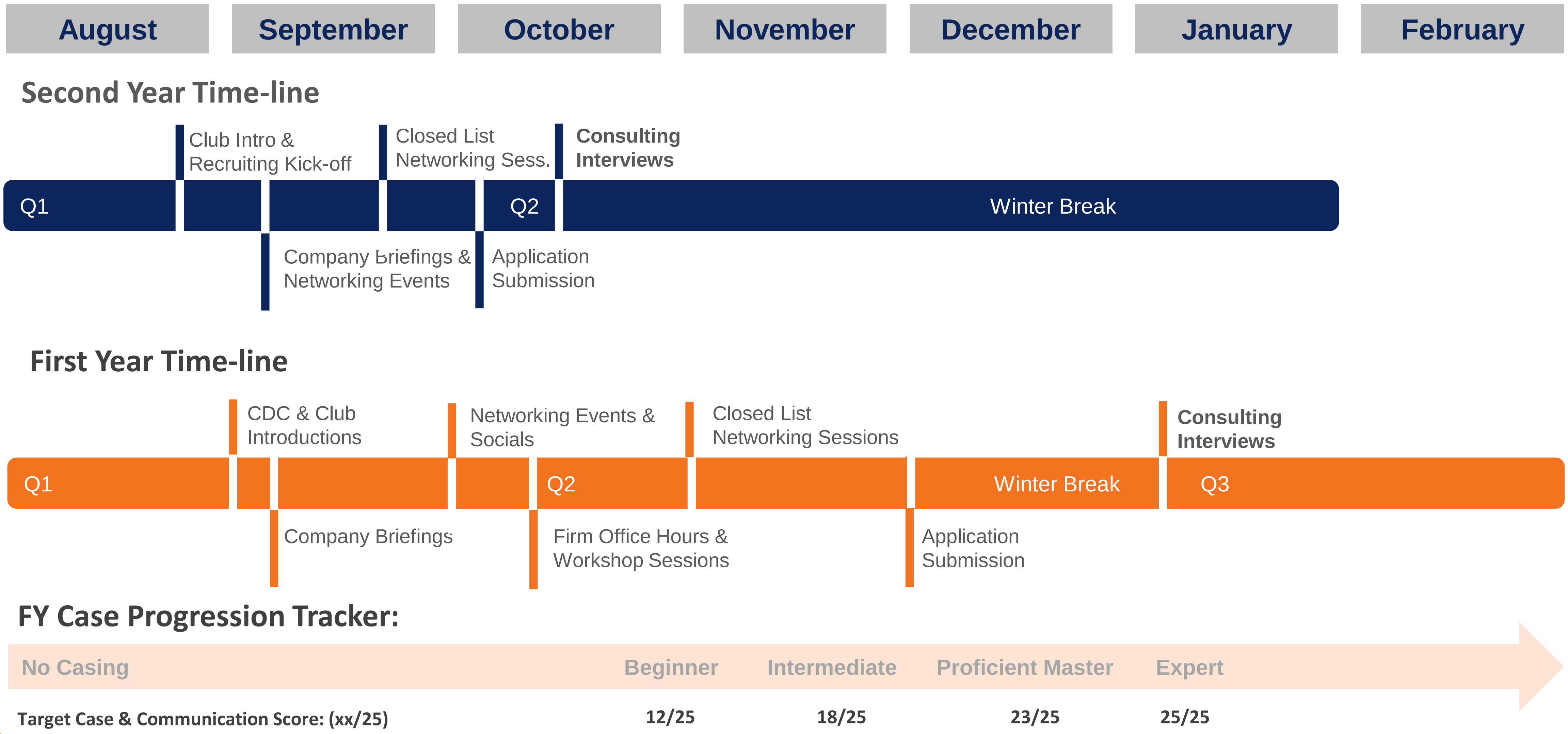
Darden Case Book 2019-2020 Index

CONSULTING CLUB AT DARDEN

*(OVR) Denotes overall case difficulty

*(N) Denotes new cases to the Darden case book (R) denotes refurbished cases

Consulting Recruiting Schedule




The Case Interview Process

Non-Verbal Communication (Tips for Success)

Communicating the correct message non-verbally is as important as the content that is communicated in a case interview. In order to recommend a candidate, interviewers must:

1. **Feel comfortable putting the candidate in front of a client**
2. **Believe they would enjoy spending 60+ hours a week with the candidate on a project team**

You Need to Convey

- Trustworthiness
  - Confidence
  - Competence
  - Honesty
- Positivity
- Curiosity
- Passion

How to Develop These Traits

- Practice and gain proficiency in the technical requirements of a case – free up mental energy to focus on connecting with the interviewer
- Practice positive self-talk – the best way to demonstrate these traits is to truly believe you have them
- Practice quickly connecting with people about non-work subjects – building rapport before diving into the interview makes an enormous difference (Note: Not always possible)

What Is A Case Interview?

A case interview is a simplified, condensed version of a complete consulting project. The candidate is in the driver's seat and is expected to explore the data and provide solutions to an issue that a client paid the firm for.

Ideal Candidates Demonstrate | **Problem-solving Ability** + **Interpersonal Skills** + **Cultural Fit & Passion**

From Bain's Website (but representative of all firms):

The case interview is an example of a real business problem based on your interviewer's past work experiences. The problems you will encounter are not designed to be brainteasers, or theoretical problems designed to stump you, but rather to reflect the challenges that our clients face. These real-life examples allow you to learn more about the type of work we do and the impact we have with our clients.

The firms look for:

- The approach you take to solving a problem
- How analytical and creative your thinking is
- Your usage of data to quantify your recommendation(s)
- Your communication skills in conveying your ideas
- How you would suggest implementing those proposals"

Consulting Case Themes

The following case themes are the most popular topics tested and they are ranked in descending order of frequency. Please note that there are plenty more case topics than the 5 listed below.

Profitability: Analyze potential sources of profit declines and identify ways to improve profitability

Market Entry / Market Sizing: Analyze the client's opportunity to expand and quantify the viable market for any new products

Growth: Identify opportunities for the client to optimally grow revenues or increase market share

Acquisition / Sale: Determine whether the client should purchase another business or sell an existing part of the company

Industry Assessment: Assess the health and attractiveness of a particular industry to advise a client's decision

Note: This not an exhaustive list of all the types of cases you may experience

Case Interview Format

Intro	Fit Question	Case Interview	Q&A
3-5 Mins	5 Mins	20-30 Mins	5 Mins

Prompt & Clarifying Q.

Prompt: This is the premise of the case. It will include some background on who the client is and lay out the problem.

Clarifying Questions: Some basic contextual questions to guide your framework; not a time to get into the nitty gritty

Framework

Framework: Once you are confident that you understand the prompt and context, you will take roughly 90 seconds to lay out a "roadmap" for how to approach the case. See the section on "Case Frameworks" for more tips.

Exhibits & Analysis

Exhibits: Provide important insights and next steps in cracking a case. You should walk through each exhibit with the interviewer, confirming understanding of its content and identifying relevant patterns or insights.

Math: See the section on "Math" for more tips.

Brain-storming

Brainstorming: After the initial analysis, the interviewer will often ask you to brainstorm around additional aspects of the problem. See the section on "Brainstorming" for more tips.

Conclusion/ Next Steps

Conclusion: Recap (briefly!) the initial problem, present the logical conclusions and solutions that arose from analysis and brainstorming (along with the data to support them.)

Next Steps: Suggest some next steps for implementation or further research and point out any potential risks

Note: This format may differ by firm or by interviewer

Fit / Behavioral Question Overview

Consulting firms are assessing your ability to successfully work with teams and judge how well you understand their firm and yourself

Tell Me About Yourself

- **Your pitch should include:** Where you have been / what you have done, who / where you are now, and what you are interested in going forward
- **All answers must include:** Relative skills & experiences, key transition points in your career, and a career objective
- **Strong answers have:** Powerful "hook" –your value proposition that highlights how you have the competencies they are recruiting for

Why Firm X or Consult-ing

- **Firm & industry fit:** Interviewee should have a deep understanding of a firm's culture, functional expertise, and working style
- **Industry interest:** Interviewee should highlight experiences they want to gain and skills they can bring
- **Strong answers have:** This question answered in opening pitch, structured reasons why, and answers tailored to your prior experiences and future aspirations

Tell Me About A Time When…

- **Experience/ behavioral:** Interviewee should demonstrate leadership ability, "drive" –taking initiative, and professionalism
- **CAR: Context / challenge** you faced, the **action** that you demonstrated, and the **result** of your actions
- **Strong answers have:** Concise response, a learning element if the situation helped you grow, utilizes structure, and engages the interviewer with energy, emotion, and authenticity

The Prompt and Clarifying Questions

Prompt & Clarifying Q. → Framework → Exhibits & Analysis → Brain-storming → Conclusion/ Next Steps

The Prompt

- All case prompts will introduce the company name, industry, and a brief description of what the client wants
- Some prompts tend to leave the primary objective to be vague
- Most prompts leave out the business context of the clients core products or operations in question

Clarifying Questions

- Clarifying questions should always be high level (ex. Context on products, core goal of the client, how client makes money)
- Don’t ask for detailed insights that you can include in framework (ex. Details on profitability drivers, market trends, etc)
- Feel free to ask further clarifying questions as you are building framework

The Case Framework

Prompt & Clarifying Q. → **Framework** → Exhibits & Analysis → Brain-storming → Conclusion/ Next Steps

By laying out a framework for analysis at the beginning of a case interview, you are not only keeping yourself organized, but you are providing a visual roadmap for the interviewer to see how you are thinking and where you are going.

Characteristics of Great Frameworks

- **MECE:** Mutually Exclusive, Collectively Exhaustive
- Detailed but not lost in the weeds
- Thorough but not wasteful
- Insightful but not presumptive

Incorrect Framework Myths

- There is one perfect framework for every case
- There is a finite number of possible frameworks that will provide answers to every case
- Frameworks are not that important to the overall interview

Framework Example

Prompt & Clarifying Q. → **Framework** → Exhibits & Analysis → Brain-storming → Conclusion/ Next Steps

Some frameworks are better than others to address a particular case. There are always multiple approaches that can yield the same answer.

A simple framework example:

Key Question in the Prompt

Notes +

Clarifying

Answers

Profitability Tree

- π
  - Revenue
    - Price
    - Volume
  - Cost
    - Fixed Cost
      - Example 1
      - Example 2
      - Example 3
    - Variable Cost
      - Example 1
      - Example 2
      - Example 3

Product Mix

Market / External

- Competitors
- Economic Health
- Regulation / Government
- Industry Trends
- Geography / Climate

Customer / Company

- Internal Capabilities
  - Acquisitions
  - Manufacturing Capacity
  - Distribution Networks
- Brand Perception
- Exit Strategies
- Consumer Preferences

Other Framework Examples

While being exposed to many frameworks can help build a strong repertoire of framework components, remember that no amount of specific framework memorization will cover every possible case a candidate could encounter.

Before & After

- Compare current with future
- Ideal for a go/no-go decision
- Can be used with profit tree (profits before and after)

	Before	After
Evaluation Criterion 1		
Evaluation Criterion 2		
Evaluation Criterion…		

M&A (similar to Before and After)

- Compare Company A, Company B, and Company A+B along various meaningful criteria

	Company A	Company B	Company A + B
Evaluation Criterion 1			
Evaluation Criterion 2			
Evaluation Criterion…			

Alternative | Evaluation Criteria Matrix)

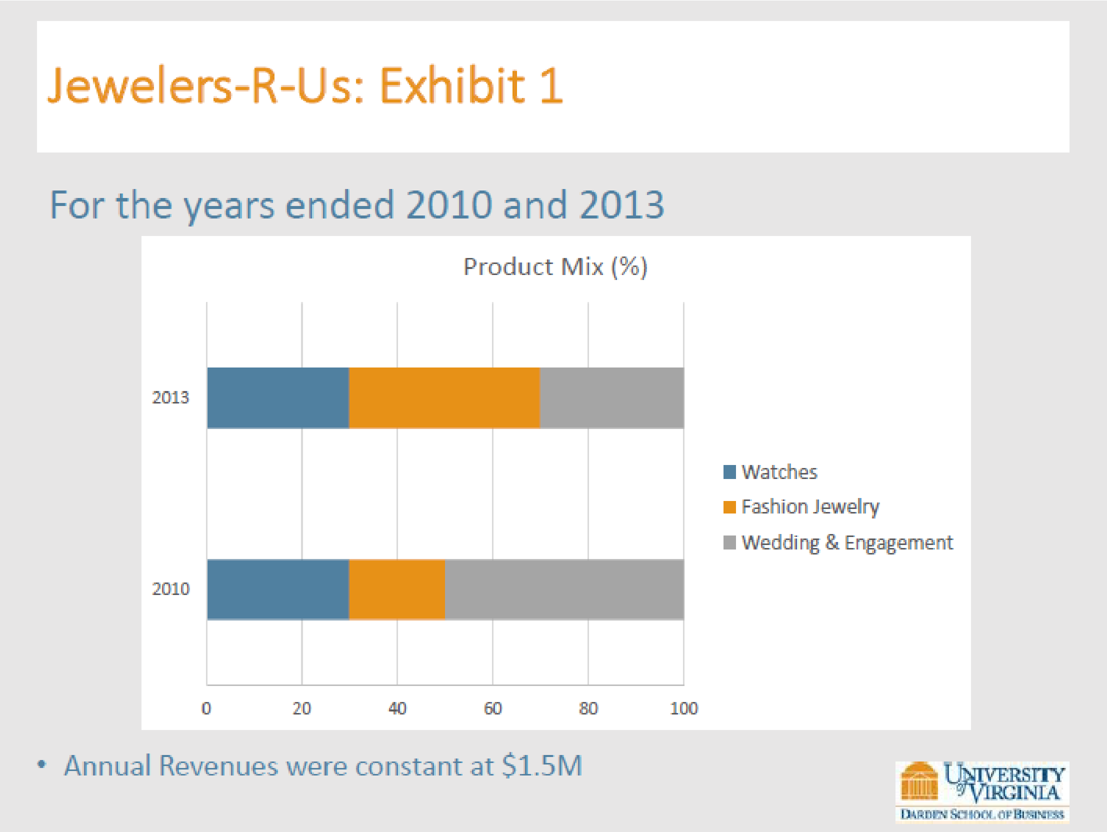
- Alternatives listed across the top
- Evaluation criteria listed along the left
- Use +, -, o, or check marks to go through the list

	Option 1	Option 2	Option 3	Option 4…
Evaluation Criterion 1	+	-	-	+
Evaluation Criterion 2	-…			
Evaluation Criteria…				

For more details on Framework, please see appendix xx

Guide to Exhibits & Analysis

Prompt & Clarifying Q. → Framework → **Exhibits & Analysis** → Brain-storming → Conclusion/ Next Steps



Read The title of the exhibit to have a clear understanding of the information

Observe The legend and gauge the sizes of any bar graphs for insights

Pay Attention To the footnotes for vital information

Guide to Math Analysis

Prompt & Clarifying Q. → Framework → **Exhibits & Analysis** → Brain-storming → Conclusion/ Next Steps

Although most case mathematics will involve simple arithmetic, it is still very easy to make mistakes. Follow the tips below to ensure that you are set up for success.

Tips for Success

Sanity Check

Make sure your numbers make sense in the context of the case

Mistakes are OK

You can recover from mistakes, just don't make the same mistake twice

Use Shortcuts

Round when needed and manage your zero's appropriately

Talk it Through

Explain each step that you are doing with your interviewer

So What

Tie the numbers back to the question and explain the implications

Practice

practice until you're comfortable doing math in front of an interviewer

Guide to Math Analysis

Prompt & Clarifying Q. → Framework → **Exhibits & Analysis** → Brain-storming → Conclusion/ Next Steps

Tricky Math Examples

Net Present Value

A. Calculates the discounted return on investment over time

B. Requires cash flows (revenues - costs over a time period), discount rate, timeframe, and an upfront cost

C. Some cases will assume a calculation into perpetuity (timeframe = forever).

D. **Perpetuity:** (Cash Flow per Period / Discount Rate) - Upfront Cost

Payback Period

A. Calculates the amount of time required to pay back an upfront investment

B. Requires cash flows per period and an upfront cost

C. Payback period will be extremely long, extremely short, or you will be given some type of specific criterion the client uses (e.g. 5 years)

D. **Payback Period** = Upfront Cost / Cash Flows per Period

Break-even Sales

A. Calculates the number of units sold to pay back an upfront investment

B. Requires per-unit selling price, per-unit costs, and upfront investment cost

C. **Break-Even Sales** = Upfront Investment Cost/(Per-Unit Selling Price - Per-Unit Costs)

Guide to Math Analysis (Things to Take Note)

Prompt & Clarifying Q. → Framework → **Exhibits & Analysis** → Brain-storming → Conclusion/ Next Steps

Example Scenario

How much money will the client save?

Current State	$ Mater. / Unit	# Labor / Unit	Volume	Total Cost
Product 1	15	10	1M	= $25M
Product 2	5	20	2M	= $50M
				= $75M

10% Labor Reduction on a per unit basis

Future State	$ Mater. / Unit	# Labor / Unit	Volume	Total Cost
Product 1	15	9	1M	= $24M
Product 2	5	18	2M	= $46M
				= $70M

Total Savings = $5M

A. **Structure** is crucial. The more you can structure information like an Excel Spreadsheet, the better
   I. Helps you recover information
   II. Your interviewer can follow along
   III. It makes you look organized

B. **Write** down prompts such as "How much money will the client save" and "10% labor reduction per unit" to keep the goal fresh in your mind

C. **Circle** any important numbers (Total Savings =$5M) that may be relevant to your final recommendation

D. **Go Deeper** by speaking to the context of the number, ex: *"I see the total savings is $5M, which represents about 6% of our total costs. This seems like a reasonable deduction but we should figure out how this labor reduction is being generated and seek to understand any negative unanticipated consequences as a result of it."*

Tips For Brainstorming

Prompt & Clarifying Q. → Framework → Exhibits & Analysis → **Brain-storming** → Conclusion/ Next Steps

The key to brainstorming is structure. The candidate should first devise two or more "buckets" or categories to organize his/her thoughts. These constraints make it easier to be creative and provide a more vivid mental model to pull from.

Example Brainstorming Frameworks:

Internal vs. External

- Internal represents any aspect internal to the client, (e.g. products, brand, financials, leadership, etc.)
- External represents factors external to the client, (e.g. market trends, geopolitical dynamic, competition, etc.)

The Supply Chain

- Each step in the supply chain is a category: raw materials, manufacturing, warehouse, transport, retail, customer, etc.
- Useful for exploring causes or solutions to cost problems

Financial vs. Non-Financial

- Financial includes things like costs, hurdle rates, cash flow, etc.
- Non-financial would be everything else

Stakeholders

- Clients, employees, investors, suppliers, regulators, etc.
- How are they affected, or what are they looking for?

Long-term vs. Short-term

- Useful for examining potential consequences of a decision

Customer Journey

- AIDA – Attention, Interest, Desire, Action
- Useful for increasing sales

Conclusion & Next Steps

Prompt & Clarifying Q. → Framework → Exhibits & Analysis → Brain-storming → **Conclusion/ Next Steps**

There are three key elements that the interviewer should include while wrapping up the final portion of the case.

1. Recommendation | Say the action that the client should take and cite the key insights in your analysis that have led you to the conclusion

2. Risks & Concerns | Mention the factors in the case that the client should be aware of that may impact the client or the recommendation –(Concerns should not contradict your overall recommendation)

3. Next Steps | Highlight at least one action the client should do to either help mitigate any concerns or any actions needed to move forward with the recommendation

While you can take a minute before you gather your recommendation, it is crucial to keep this segment concise

Industry Overview

Please note that these are commonly tested industries; This list is not exhaustive of all the industries tested

Industry Overview – Consumer / Retail

Key Industry Trends

- **Digital Marketing:** CPG (Consumer Packaged Goods) companies are pivoting to digital marketing solutions like Facebook and YouTube more than ever for smarter and more targeted advertising.
- **Big Data:** Consumer companies & retailers are ramping up the use of consumer shopping behavior data now more than ever to create curated/ personalized shopping experiences and targeted advertisements
- **Retail Omnichannel:** Large brick & Mortar retailers are pivoting to an "order online, pick-up in store" mix while also building out their online fulfillment capabilities to cater to the consumer. and keep up with Amazon). Store foot-prints are also getting smaller to reduce inventory
- **Private Label & Amazon Effect:** Private label consumer products are eroding market share of large name brand products. This is partially driven by "the Amazon effect" of quick and cheap replacement fulfillments. -Brand loyalty is getting harder and harder to win.
- **Direct to Consumer vs. In-store Experience:** Brand names are slowly shifting resources to sell directly to consumers as some retailers struggle. Retailers with large brick & mortar footprints are focusing on in-store experiences to attract customers

Important Terminology

- **SKU:** Stock Keeping Unit – Refers to a unique item sold in a store
- **In-stock:** Percent of items that are on the shelves and available for sale vs. what the total display can hold
- **CRM:** Customer Relationship Management: Strategy & tools designed to boost profitability and strengthen customer loyalty by using data – also the name for software that facilitates this
- **Loss Leader:** Merchandise sold at a loss to attract new customers or stimulate other profitable sales
- **Mark-up:** Percentage added to the cost of product to get selling price

Important Calculations

1. Inventory Turnover:

= (Sales / Inventory)

2. Gross Margin:

$$= \frac{\text{(Revenues – COGS)}}{\text{Revenues}}$$

3. Contribution Margin (CM):

= (Sales – Variable Costs)

$$\text{CM Rate} = \frac{\text{(CM)}}{\text{(Sales)}}$$

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Energy

Key Industry Trends

- **Clean Renewable Energy:** Wind, solar, and biomass power are increasingly replacing the use of fossil fuels in developed and developing countries with some projections indicating 80% of the worlds energy needs being met by renewable energy by 2050
- **Technology:** Advancements in drilling techniques like "fracking" and horizontal drilling have significantly boosted the output of US oil companies and substantially reduced the cost and risks associated with drilling for oil
- **Shale:** Newly found abundance of shale basins in the USA has helped to boost US oil production output and has almost eliminated US dependence on foreign oil
- **Natural Gas:** Given its cheap and abundant supply, natural gas has become the primary source of energy in the US, replacing crude oil and coal
- **(Important) Petroleum Products:** Gasoline, jet fuel, natural gas, fertilizer, plastics, detergent, propane, diesel, lubricant

Important Terminology

- **Upstream (E&P):** Exploration and Production – Process involving the finding, drilling, and producing of crude oil and natural gas or liquified natural gas (LNG)
- **Midstream:** Focuses on the processing, storage, marketing, and transportation of oil and natural gas. (Most pipe-line companies fall in this category)
- **Downstream:** Includes oil refineries, petrochemical plants, petroleum products distributors, retail outlets and natural gas distribution companies
- **OPEC:** Organization of Petroleum Exporting Countries – Cartel of 14 nations that coordinate petroleum policies. –Often influences output and thus oil prices

Important Calculations

1. Return on Investment (ROI)

$$= \frac{\text{(Profits – Cost of Investment)}}{\text{Cost of Investment}}$$

2. Breakeven Point

$$= \frac{\text{(Fixed Costs)}}{\text{Contribution Margin (CM)}}$$

Important Considerations:

- **Transportation / Distribution costs**
- **Storage Costs**
- **Production Costs: Labor + Materials**
- **Plant Development Costs**
- **Depreciation & Taxes**
- **Overhead**

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Transportation

Key Industry Trends

- **Airline Capacity Additions:** Airline ticket prices have been steadily on the decline driven by companies adding more routes to cities across the globe –led by low-cost carriers such as JetBlue, Southwest
- **Fuel Efficiency:** Airline companies have been investing heavily in upgrading their fleet to more fuel efficient aircrafts to reduce their biggest cost driver
- **EV (Electric Vehicles):** Auto manufacturers are all racing to create battery powered vehicles
- **Autonomous Vehicles:** Autonomous vehicles are expected to hit the road as early as 2019 and will cause major disruption to auto manufacturers, bus systems, taxis, insurance companies
- **Shortage of Truckers:** Transportation companies have been struggling to keep up with the booming demand for cargo shipments due to a massive shortage of truck drivers –thus causing significant increases in labor costs

Important Terminology

- **Load Factor:** Measures the capacity utilization of transportation services and is equal to the average actual utilization divided by the maximum capacity
- **PRASM: Passenger Revenue per Average Seat Mile –**Or RASM (revenue) is the revenue generated per available set miles in which ASM = number of seats available x number of miles flown.
- **Logistics:** The detailed coordination of complex operations involving many people, facilities, or supplies.
- **FOB: Free On Board** –Represents the point at which the sale of a freight cargo is considered complete. “FOB shipping” means ownership is transferred once the product is shipped of, “FOB shipping point” means ownership is transferred once the product is delivered
- **LTL & FTL:** LTL (Less than Load) – Small freight that doesn’t fill a truck which is generally more expensive to ship, (FTL) Full Truck Load) – Large shipments that fill a trailer and are thus cheaper to ship

Important Calculations

1. Potential Savings by Switching Equipment

= {New Profit – Old Profit} or

{ [(New Capacity x Price) – (New efficiency x cost)] – [(Old Capacity x Price) – (Old efficiency x cost)]}

Important Considerations:

- **Gasoline / Fuel Prices**
- **Carrying Capacity**
- **Range / Distance**
- **Destination Routes**
- **Maintenance Costs**
- **Depreciation**

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Manufacturing / Agriculture

Key Industry Trends

- **D2C: Direct to Consumer:** More manufacturers are leveraging their own sales platform to market, sell, and ship their products to the customer rather than use third party distributers or retailers to boost profitability
- **Data Driven Analytics:** Manufacturers are using predictive analytics and algorithms to improve product design, optimize production cycles, and improve demand forecasting
- **Trade-war & Tariffs:** In response to the US tariff on steel and more, Canada, the European Union, and China have all implemented retaliatory tariffs of close to 25% on agricultural and automobile goods produced in the US.
- **Sustainable Food Systems:** Vertical farming has been a growing trend in urban locations to minimize environmental foot-prints and bring produce to major cities

Important Terminology

- **(JIT) Just-in Time Inventory:** "Pull demand" inventory system in which assembly materials and support items are delivered as needed to minimize raw material inventory
- **Commodity:** An interchangeable non-differentiated product or material that is sold freely. (Most agricultural products are commodities)
- **Bottleneck:** The resource in a manufacturing process that is working at max capacity and thus limits the output of the entire production
- **Bushel:** A unit of dry measure (1 cubic foot) for grain, fruit, etc., equivalent to 8 gallons of liquid
- **Out-source:** Process of contracting an outside party to complete a production or service task for a business. –Typically done to save cost or due to a lack of expertise

Important Calculations

1. Potential Savings with New Equipment

= (New Equip. Expenses – Old Equip. Expenses)

[(Old Time x Old Labor) + (Raw Material Cost x Old Quantity) + Old Depreciation)]

- [(New Time x New Labor) + (Raw Material Cost x New Quantity) + New Depreciation)

Important Considerations:

- Raw Material Costs
- Labor & Wages
- Capacity Constraints / Bottlenecks
- Commodity or Not?
- Overhead Costs
- Supplier & Buyer Relationships
- Depreciation

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Financial Services

Key Industry Trends

- **AI, Block-Chain & Crypto Currencies:** Digital distributed ledgers offer a cheaper and more efficient way for firms to verify and facilitate transactions. Crypto currencies have proven themselves to be an alternative set of asset investments that rival equities, precious metals, and debt holdings
- **Digital-Only Banks & Payments:** The prevalence of more digital transactions have eroded the need for cash for most daily use, which has in turn lead to the proliferation of online banks that offer higher savings account interest rates and comparable services
- **Financial De-regulation:** Congress passed legislation easing some of the restrictions from Dodd- Frank that exempts smaller banks from certain capital requirements which frees up room for more loans
- **More Transparency in PE Funds:** With greater pressure to produce results that outperform their benchmarks, more PE investors have been demanding greater transparency within their funds and firms have been using transparency to attract investors

Important Terminology

- **AUM:** Assets Under Management: Market value of all the financial assets that a firm manages on behalf of all of their clients and themselves. –Includes capital raised by investors and leaders of a firm
- **Private Equity:** Composed of investors and funds that invest directly into private companies or convert public companies to private companies to improve the target company’s operations and financials with the goal of extracting a financial return from the company and reselling it another firm or the public at a profit
- **M&A:** Mergers & Acquisition: Mergers are when two companies comes together to make a new entity (Dow Chemical & Dupont) = DowDuPont, while an acquisition is where the smaller company is consumed by the larger company (Amazon + Wholefoods) = Amazon

Important Calculations

1. NPV (Net Present Value)

$$= (CF) \times \frac{1}{(1+i)^n}$$

Where n = # of periods

2. Pay Back Period

$$= \frac{\text{(Fixed Costs)}}{\text{Contribution Margin (CM)}}$$

Important Considerations:

- **Current Portfolio**
- **Exit Strategy & Time Horizon**
- **Acquisition Price**
- **Employee & Customer Relationships**
- **Market Trends**
- **Tax & Regulatory Implications**
- **Client Risk Profile**

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Information Technology

Key Industry Trends

- **Artificial Intelligence (AI)/ Machine Learning:** Artificial intelligence is the ability for a computer program to think and learn. The emergence of AI has enabled the rise of self-driving cars, smart homes, advanced search algorithms, and smart digital assistants
- **Cloud Computing:** Is the practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer. More companies are moving to this platform for security, convenience, and cost savings
- **Internet of Things (IOT):** Smart devices that are all connected and communicate with each other via the internet are rising in demand due to value of strategic data that they provide
- **Blockchain:** a digital ledger in which transactions made and recorded chronologically and publicly. – Important for security and transfer verification purposes. Ex. include Bitcoin, and other cryptocurrencies
- **GDPR:** General Data Protection Regulation: Data protection regulation protecting privacy for all individuals in the European Union.

Important Terminology

- **IP (Intellectual Property):** A category of property that includes intangible creations protected by trademarks and copyrights (e.g. software, code, algorithms, etc.)
- **Unicorn:** a start-up company valued at more than a billion dollars, typically in the software or technology sector
- **Freemium:** A pricing model used by many digital services, a "freemium" model is one where the majority of users are able to engage with a product or service entirely for free (perhaps in exchange for data collection or being served advertisements)
- **SaaS:** "Software as a service" - a software distribution model in which a third-party provider hosts applications and makes them available to customers over the Internet –Like Salesforce or Workday

Important Calculations

1. Addressable Market size:

Top-Down: Total Population >>> Number of users >>> Market share >>> # of Units per User x Price per Unit

Bottom-Up: Current Customer Population >>> Potential Customer Base (Estimated using consensus data or industry info) >>> Future user base x units per user x price

2. Customer Acquisition Cost:

$$\frac{\text{Marketing Expenses}}{\text{Newly Acquired Customers (Yearly)}}$$

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Media & Entertainment

Key Industry Trends

- **Cord Cutting / Over the Top Streaming:** The rise of Hulu, Netflix, YouTube, & Amazon Prime video has left many to abandon traditional cable and opt for online streaming services to get the content they want
- **Content is King:** Media giants have been spending heavily to curate high quality content to hook subscribers to their service and maintain and grow their subscriber base
- **Ad-model Shift:** Cable advertisement has been trending downward while digital online advertisements have been trending up. As online viewers opt for ad-blockers, AI and big data are helping marketing agencies personalize advertisements and increase user engagement
- **Augmented Realty (AR)/ Virtual Reality (VR):** While still in their early stages, AR and VR capabilities have been gaining traction in the industry as a way to enhance storytelling and improve sporting coverage
- **Music Streaming:** The rise of Spotify, Apple Music, & YouTube Music has almost eliminated the physical disc music market as most artist now prioritize online platforms to release albums and new songs
- **Gaming & E-Sports:** The video gaming industry has been one of the fastest growing segments in entertainment led by mobile gaming and game streaming experiences via Twitch and E-sports. Many video game creators are focused on a "games as a service model" as they monetize video games overtime by selling in-game customizable perks

Important Terminology

- **Digital vs. Linear:** Linear is traditional broadcast or cable television. Digital is online (streaming, etc.)
- **Ratings:** A measure of viewers of a particular program or time segment in television. Nielsen is the largest provider of ratings data in the US, but has been slow to provide digital ratings
- **Box-Office:** The total revenue generated by movies shown at theaters

Important Calculations

1. Profitability

(Revenues – Costs)

(Price x Quantity) – (Quantity x Var. Cost) – (Fixed Costs)

Important Considerations:

- Revenue Factors
  - Advertising Rev.
  - Ticket sales (Price x Quantity)
  - Merchandising
  - Tours / licensing / Endorsements
- Cost Factors
  - Artist fees
  - Commission
  - Promotion advertising
  - Venues
  - Content creation costs

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Healthcare & Life Sciences

Key Industry Trends

- **Wearable Medical Devices:** Activity trackers help patients stay more active and healthier on their own while also monitoring health metrics reducing the need to visit doctors frequently
- **Smart Technology & Data:** Data on a patient's background and conditions allow more personalization options, targeted treatments, and faster recommendations at hospitals
- **Gene Therapy:** The transplantation of normal genes into cells in place of missing or defective ones in order to correct genetic disorders. –Growing trend using CRISPR to treat previously uncurable diseases
- **Price Transparency:** As drug companies receive criticism on the rising cost of their drugs, more states are considering independent efforts to improve transparency in drug pricing and cost controls
- **Government:** With the repeal of the Affordable Care Act (ACA), legislation has stagnated on fixing the rising cost of healthcare and Medicaid in the US, thus leaving a continued rise in insurance premiums
- **Bundled payment, episode-of-care payment, etc.:** Generally describes paying for the whole treatment at once, rather than by individual tests or visits – an attempt to incentivize improved outcomes

Important Terminology

- **Orphan Drug:** A pharmaceutical drug that remains commercially undeveloped due limited potential for profitability as a result of a small curable population size
- **FDA:** "Food & Drug Administration" Federal organization tasked with protecting and promoting the safety of food and pharmaceuticals in the US. FDA approval is needed for almost all drugs sold in the US
- **Generic Drugs:** A prescription drug that has the same active-ingredient formula as a brand-name drug but sold at a cheaper cost. –Typically occurs when name branded drugs lose patents
- **Biotech vs. Pharmaceutical:** Biotech firms use live organisms like bacteria and enzymes to manufacture their medicines while pharmaceutical companies primarily use chemicals synthesis
- **Auto-immune Diseases:** A disease in which the body's immune system attacks healthy cells

Important Calculations

1. Market sizing:

Top-Down: Total Population >>> Number with Illness >>> Number Diagnosed >>> Market share of Drug >>>> (Dosage per Time Frame) x Price per Dosage = Market Size per Time Frame

Important Considerations:

- Regulations
  - FDA Approvals
  - Patent Rights
  - Foreign Government Laws
- Competition / Cannibalization
- Drug Effectiveness
  - Cure vs. Treatment
  - Time to Market
  - Side Effects
- Manufacturing Capabilities
- Pricing, Costs (Fixed / Var.), Dosage

*Please note that not all trends, terminologies, and calculations are listed above

Industry Overview – Telecommunications

Key Industry Trends

- **5G Network Service:** Next generation of mobile internet connectivity with faster speeds, more reliable connections, and 100x more bandwidth capacity than 4G.
  - Network operates mainly on the cloud
  - Allows for “network slicing:” Creates separate wireless networks on the cloud for users to have their own personalized network
  - Roll-out may be 2020 in North America due to high infrastructure costs associated with development
- **Network Consolidation:** The third and fourth largest cell phone carriers T-Mobile and Sprint are in the process of merging, a move that will consolidate the telecom market to 3 major players
- **Content Integration:** High profile acquisition like AT&T of Time Warner and Verizon of Yahoo illustrate a push to either get into the content creation game or to build out their advertising network

Important Terminology

- **Carrier:** A company that is authorized by regulatory agencies to operate a telecommunications service system: AT&T, Verizon, T-Mobile
- **OEM:** Original Equipment Manufacturer – A company whose goods are used as components in the product of another company that sells the finished goods to users
- **LAN:** Local Area Network: Locally owned and administered data network that runs primarily through cables –ex. Ethernet connection
- **Fiber Optic:** Transmission connectivity via glass strands which are 100x more faster than traditional copper wires for more efficient cell phone and internet connections

Important Calculations

1. Return on Investment (ROI):

$$\frac{\text{(Future Profits – Cost of Investment)}}{\text{(Cost of Investment)}}$$

2. Customer Acquisition Cost:

$$\frac{\text{Marketing Expenses}}{\text{Newly Acquired Customers (Yearly)}}$$

Important Considerations:

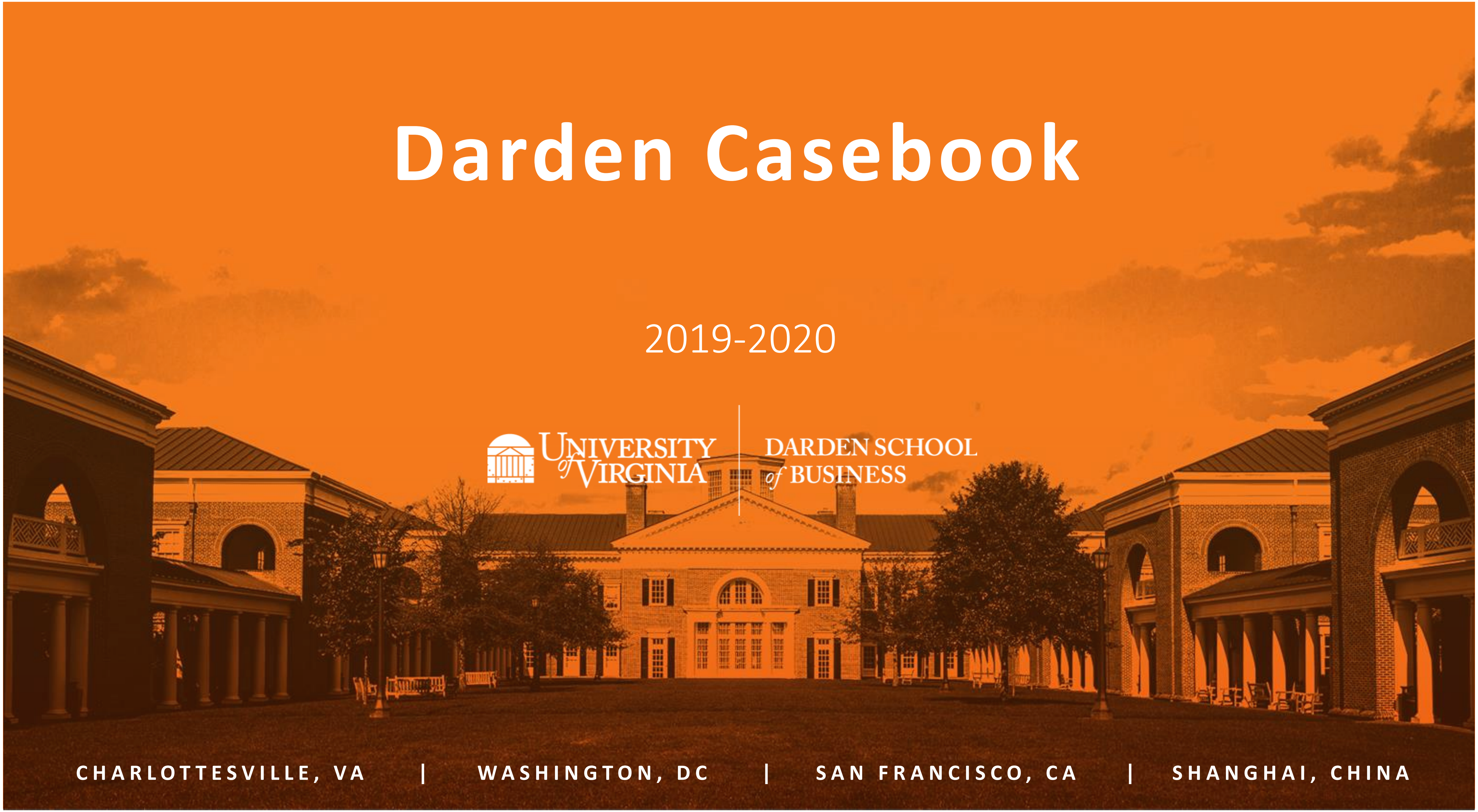
- Regional Competition
- Competitors
  - New Entrants
  - Barriers to Entry
  - Substitutability
- Contract lengths & stipulations
- Infrastructure

*Please note that not all trends, terminologies, and calculations are listed above

Darden Casebook

2019-2020

UNIVERSITY of VIRGINIA | DARDEN SCHOOL of BUSINESS



CHARLOTTESVILLE, VA | WASHINGTON, DC | SAN FRANCISCO, CA | SHANGHAI, CHINA

Darden Case Book 2019-2020 Index

Case Title		Industry	Case Type	Difficulty Quant / Qual / OVR*			Page
Gamebox	N*	Entertainment	Product Launch	2	1	1	35
Seven Flags	R*	Hospitality	Pricing	2	1	1	45
Smiles4Life	N	Healthcare	Growth Strategy	2	2	2	55
Digging for Gold	N	Mining	Market Entry	2	2	2	66
PharmaCo	N	Pharmaceutical	M&A	2	2	2	74
Sisha: Just Blowing Smoke?	N	Public Sector	Market Entry	2	2	2	82
MBS Co.	N	Financial Services	Profitability / ROC	2	3	3	92
Emerald City	N	Public Sector	Other (FDI)	2	3	3	104
Omega Naval Systems	N	Aero & Defense	Profitability	3	2	3	112
Boxing in the Opportunity	N	Private Equity	M&A	3	2	3	123
To Automate or Not	R	Logistics	Investment Decision	2	3	3	133
Rubber Bumper	R	Manufacturing	Profitability	3	3	3	144

*(OVR) Denotes overall case difficulty

*(N) Denotes new cases to the Darden case book (R) denotes refurbished cases

To get the most authentic casing experience, you should aim to do at least one behavioral interview question at the start of each case and have your interviewer take note of your timing

Indicates the overall difficulty of the case with a combination of mathematical technicality and creative thinking ability. It is advised that you start with single star cases and work your way up to three star cases

Indicates the degree of math difficulty in the case. Harder cases typically have multiple stages of calculations with multiple opportunities for mistakes

Indicates how creatively intensive the case will be. The more qualitative, the more thorough your framework should be. These cases will place more emphasis on the brainstorming elements

Grading Rubric

Case Execution	Communication	Behavioral
High scorers should be well structured, demonstrate coachability, and make insightful connections	High scorers should demonstrate confidence, speak clearly, and have a tidy case work	High scorers should give clear and concise answers that are relevant

Gamebox

Gaming/Entertainment

GAMEBOX

Gaming/Entertainment

Prompt:
The CEO of GameBox, a well-known video game company, has approached us to help decide what type of new game the company should develop. How would you advise her?

Clarifying Information: *Note: Provide this only if corresponding questions are asked.*

What is the CEO's main goal?
To maximize profit.

How does Gamebox make money?
Gamebox makes both its own games and consoles (similar to Nintendo). They make money by selling new consoles and video games. They are the sole producer of Gamebox games. Games are almost entirely sold digitally on Gamebox's online portal.

What does the CEO mean by "type" of game?
Game Type relates to the category of game. For example, sports, family, racing, etc.

Are we concerned about the game's age rating?
Most Gamebox games are rated E for Everyone or T for Teen. Gamebox has never produced a Mature rated game.

BEHAVIORAL INTERVIEW QUESTION:

1. What is your favorite class at Darden and why?

2. Tell me about a time where you had to make a decision with limited information. What was your approach?

Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions.

Financial

Revenues

- Expected # of game copies sold
- Price point of game
- Create incremental demand for consoles
- Alternative revenue sources
  - Merchandise
  - Add-ons/In-game purchases
  - Partnerships

Costs

- Game Development
  - Software developer salary
  - Storyboarding
  - Equipment, licenses, and other overhead
  - Testing
  - Intellectual property
- Marketing/Awareness
- Marginal cost of each game sold

Non-Financial

Internal

- Existing game portfolio. Saturation in any category?
- Expertise by game type
- Customers
  - Preferences
  - Willingness to pay
- Fit with mission/long-term strategy of Gamebox

External

- Market-wide trends by game type
- Trend to mobile gaming
- Alt. forms of entertainment (movies, live action, etc.)
- Disposable income
- Game pipeline of major competitors

How to Move Forward:

The candidate should recognize this as a profitability case, but one that must first identify and size the market. Interviewer should push the candidate to first tackle the market sizing problem. Provide Exhibit 1 only after the candidate has brainstormed ways to segment and size the market (i.e. age, gender, game category preferences, etc.)

EXHIBIT 1

Video game market trends

Game Category	Age < 18	Age 18-30	Age 31+
Sports	43%	60%	20%
Shooter	35%	38%	25%
Racing	12%	35%	28%
Fantasy	15%	23%	22%
# of gamers with GameBox console (millions)	8	20	15

Question 1

- What category of game do you recommend our client pursue?

Supply only when asked:

- Price point of all games will be $50, regardless of category
- Development cost of all games is $50 Million, regardless of category
- Marginal cost of selling a copy of any game is negligible since games are sold digitally

Exhibit or Question Guidance:

After looking at Exhibit 1, candidates should be able to identify sports games as having the highest potential market

- Because any age group can own multiple types of games, percentages sum to greater than 100%
- Candidate should ask to round numbers. If they do not, interviewer should recommend to do so. Figures below are exact
- Strong candidates will:
  - Calculate revenue potential of selling games after calculating total customers
  - Inquire about other revenue sources. If not, prompt him/her to brainstorm additional ways to monetize the game

All figures in millions	<18	18-30	31+	Total Customers	Total Revenue
Sports	3.44	12.00	3.00	18.44	922
Shooter	2.80	7.60	3.75	14.15	708
Racing	0.96	7.00	4.20	12.16	608
Fantasy	1.20	4.60	3.30	9.10	455

BRAINSTORMING

What are other ways Gamebox could monetize revenue for their games?

Brainstorming Guidance:

Note: This is just one possible set of categories and answers. Many more are possible, and interviewers should assess both the volume and relevance of answers.

Internal Products

- Merchandise
- In-game add-ons
- Game sequels
- Premium services
- Special/Limited edition version
- Game score music album
- Bundled packages

External Partnerships

- License IP for music, movies, television
- Theme park events
- Tournaments/Competitions
- Selling usage data to interested parties
- Youtube videos that run ads

Best candidates display:

After the candidate has produced an initial list, push him/her to think of a few more ideas. Once an exhaustive list is generated, great candidates will take the initiative to ask interviewer if he/she has any relevant information.

EXHIBIT 2

Add-on Purchase Interest

Game Category	# of Purchase Interests*	Avg. Price of Add-On**
Sports	5	$10
Shooter	9	$30
Racing	6	$30
Fantasy	7	$50

**In millions of users*

***The term add-on refers to any purchased expansionary pack or supplement that is added to an existing game.*

Question 2

- Has your recommendation changed after viewing this information?

Exhibit or Question Guidance:

Clarifying questions:

- Are there costs associated with add-ons? Yes, but they are negligible in comparison to the selling price of the add-ons.
- Each purchase interest equates to one add-on purchase. Figures are estimates from the marketing department based on historical consumer behavior.

Evaluation:

- Candidate should work out the revenue associated with add-ons
- Candidates should point out that the “Fantasy” game category produces the most revenue from add-ons. Good candidates will immediately combine revenues from games sold + add-ons. Doing so concludes that the “Shooter” game category produces the most value from revenue.
- Interviewee must remember to include development cost of $50M.

Game Category	Purchase Interests	Avg Price	Add-On Revenue	Games Sold Revenue	Total Revenue	Total Profit**
Sports	5	$ 10	50	922	972	922
Shooter	9	$ 35	315	708	1,023	973
Racing	6	$ 30	180	608	788	738
Fantasy	7	$ 50	350	455	805	755

**All figures in millions except for Avg. Price*

***Includes $50 M of development cost*

CONCLUSION

To conclude, the interviewee should provide the following:

Recommendation:

- Candidate should give a clear, succinct recommendation about which game category to pursue. Candidate should use data from the case to support his/her recommendation.

Risks:

- Add-on revenue is sensitive to total game sales and (potentially) changed direction of recommendation
- (If Shooter) Given GameBox's primary demographic, how the game is rated may impact sales

Next Steps:

- Quantify opportunity of alternative revenue sources generated in brainstorming session to gain confidence in recommendation
- Develop go-to-market strategy from storyboarding to development to launch

INTERVIEWER FEEDBACK FORM

Case Name ______________________ **Interviewer** ______________________

Case Book ________________ **Case Type** __________ **Difficulty** __________

Case Execution:

- ❑ Clarifying Questions + Framework
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

1 2 3 4 5

Feedback:

- ❑ Exhibits + Quantitative Ability
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

1 2 3 4 5

Feedback:

- ❑ Brainstorm + Conclusion
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

1 2 3 4 5

Feedback:

- ❑ Presence & Non-Verbal
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

1 2 3 4 5

Feedback:

Total: _____ / 20

Seven Flags

Hospitality

SEVEN FLAGS

Hospitality

Prompt: Our client is a mid-size amusement park chain, with 10 parks around the U.S. serving over 10 million visitors each year. In their Richmond, VA park, it operates both a traditional thrill-ride section, as well as an animal experience. **(Show park map.)** Currently, the two sections are covered under one entry ticket price. However, management is considering offering a separate ticket for only the animal experience section. They have come to us to determine if this is a good idea.

Clarifying Information: *Note: Provide this only if corresponding questions are asked.*

1. Financial goal: Management wants a payback period less than 10 years. (If the candidate asks, payback period = investment / on-going profit.)

2. Current price: Tickets are currently $20 and provide visitors full access to the park

3. Park attendance: The Richmond, VA park is an average sized park within the client's portfolio

4. Business model: The park is a typical amusement park (think Six Flags or Busch Gardens). Visitors buy a ticket for entrance (assume same price for adults and children), and all rides / amusements are accessible under the one ticket price. The park also sells merchandise and food / drinks separately.

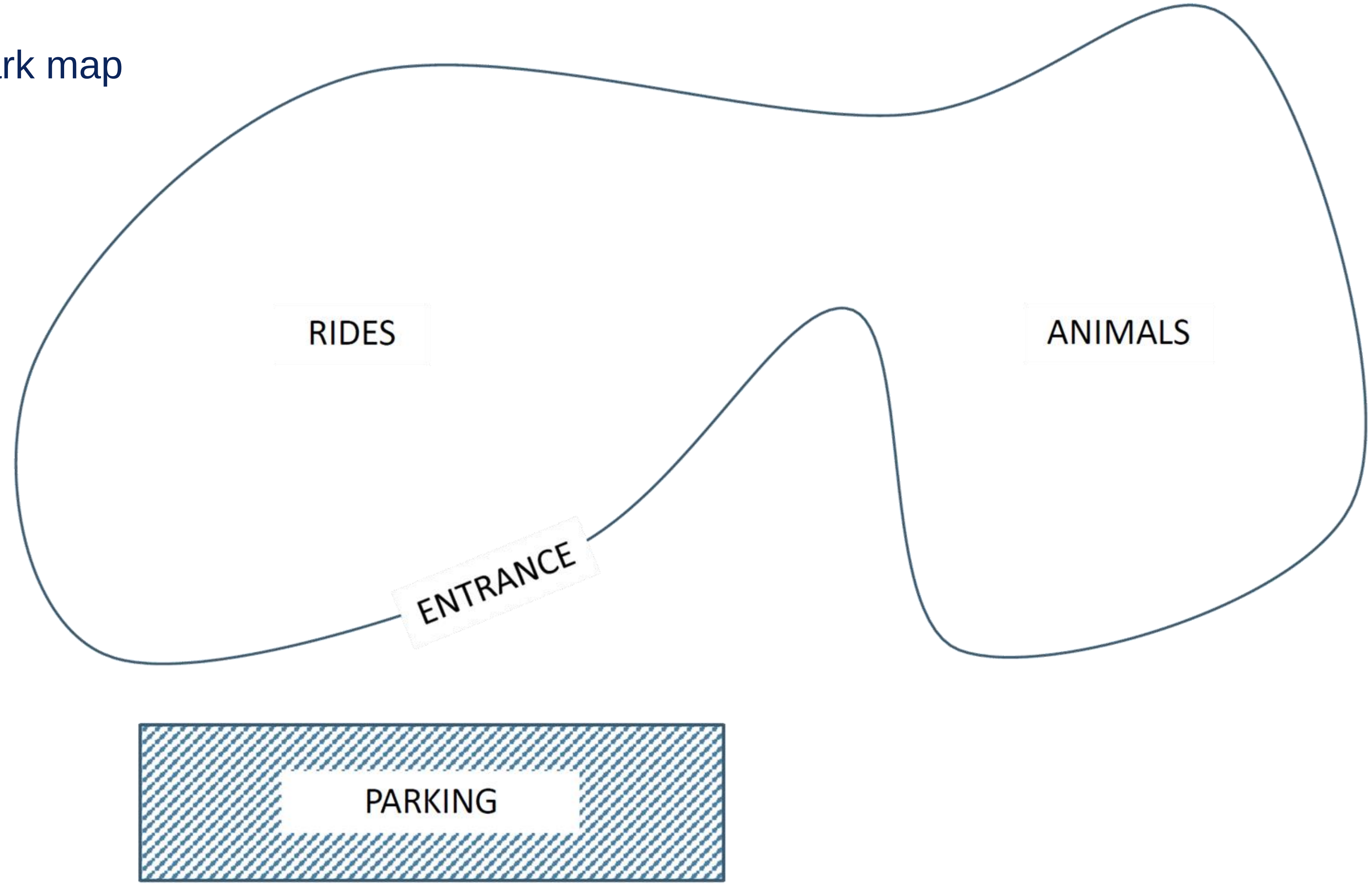
BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about yourself.

2. Tell me about a time you helped a team overcome a problem.

EXHIBIT 1

Seven Flags park map



Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions.

Incremental profit

- Revenue: ticket prices, food & drink sales, merchandise, visitor volume, cannibalization of "ride + animal" ticket sales
- Existing costs: maintenance, animal care, labor/operations, COGS (merchandise, food)
- New costs: Build out of new entrance, wall between sections of the park, and new parking

Competitive landscape

- Other attractions in the area
- Zoos, petting zoos
- Museums
- Movie theaters
- State fairs
- All forms of family friendly recreational activities

Macro trends

- Changes in disposable income
- Weather conditions
- Consumer entertainment preferences

How to Move Forward:

Key insights include pricing decision, cannibalization impact, and recovery of new fixed costs. If candidate does not identify these in his/her framework, push candidate to brainstorm incremental changes in revenue and costs.

When asked about pricing, ask candidate how he/she would determine price. After the candidate has brainstormed some ideas, present Exhibit 2.

EXHIBIT 1

Price elasticity of animal only admissions

1. Cannibalization rate is 50%
2. Establishment is open 350 days per year

Question 1 From Exhibit 2, candidate should want to discover which price would maximize revenue. See below for solution.

Exhibit or Question Guidance:

Ticket price	Number of Visitors	Cannibalization rate	New tickets	New ticket revenue	Cannibalized tickets	Cannibalized revenue per ticket (current ticket $20)	Daily canibalized revenue	Daily incremental revenue	Yearly incremental revenue
A	B	C	D = B * (1-C)	E = A * D	F = B * C	G = A - $20	H = G * F	I = H + E	J = I * 350
$10	1,300	50%	650	$6,500	650	($10)	($6,500)	$-	$-
$12	1,100	50%	550	$6,600	550	($8)	($4,400)	$2,200	$770,000
$14	700	50%	350	$4,900	350	($6)	($2,100)	$2,800	$980,000
$16	500	50%	250	$4,000	250	($4)	($1,000)	**$3,000**	**$1,050,000**

- Revenue is maximized at a price point of $16 per ticket for the animal only admission. This equates to ~$1MM in incremental ticket sales or a 5% increase
  - From the prompt and clarifying questions, we know that Seven Flags sees about a million visitors a year at $20 per visitor in ticket sales. Thus, current ticket revenue equals $20MM.
- Great candidates will mention that incremental revenue is sensitive to the cannibalization rate assumption

BRAINSTORMING

Aside from ticket sales, what are some other considerations that will factor into the decision?

Brainstorming Guidance:

Note: This is just one possible set of categories and answers. Many more are possible, and interviewers should assess both the volume and relevance of answers.

Other revenue sources:

- Food / beverages
- Merchandise
- Parking
- Annual passes (individual / family)

Incremental costs:

- Build out of new entrance, building a wall to separate rides from animal enclosure, and additional parking
- Additional employees to serve higher visitor volume
- Payback period less than 10 yrs?

Best candidates display:

Great candidates will have asked about specific KPI's at the beginning of the case (i.e. projects must have a 10-yr. payback period or better). Candidates should remember on his/her own to evaluate the payback period of this project and ask for the relevant information.

This brainstorming activity is an opportunity for great candidates to leverage their framework. Great candidates will return to their framework to recall the primary objective and other ideas already generated.

Question 2 What is the payback period of this proposal?

- Great candidates should recognize that they need to do this calculation. If not, help the candidate recall that one metric Seven Flags uses to evaluate projects is a 10-yr. payback period.
- Provide only when asked: Fixed costs for constructing new entrance, wall, and parking lot is estimated to be $2 million.
- Only consider incremental revenue from ticket sales
- Seven Flags profit margin in 20%

Exhibit or Question Guidance:

- Candidate only needs to calculate the payback period at the $16 price point
- Payback period = investment / incremental profit
  - Incremental profit = incremental revenue x profit margin. Incremental profit = $1,050,000 x 20% = $210,000
  - Payback period = $2,000,000 / $210,000 = ~9.5 years

CONCLUSION

To conclude, the interviewee should provide the following:

Recommendation: There is no correct answer, but a likely response could be:

- Move forward with creating an animal only admission ticket
- Incremental revenue / profit is maximized at a price point of $16.00 per ticket, leading to a 5% increase in ticket revenues and profits
- Given the $2M investment, payback period is ~9.5 years – just below management's requirement of 10 years

Risks:

- Incremental revenue / profit very sensitive to cannibalization. If actual cannibalization is higher, incremental profits will suffer and payback period will exceed 10 years
- Potential opportunity cost to invest in higher ROI projects

Next Steps:

- Payback period will decrease (improve) if animal park can sell additional merchandise, food, and beverages to new visitors. Size this opportunity and understand impact

INTERVIEWER FEEDBACK FORM

Case Name ______________________________ **Interviewer** ________________________________

Case Book ________________________ **Case Type** ______________ **Difficulty** ______________

Case Execution:

- ❑ Clarifying Questions + Framework
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

1 2 3 4 5

Feedback:

- ❑ Exhibits + Quantitative Ability
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

1 2 3 4 5

Feedback:

- ❑ Brainstorm + Conclusion
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

1 2 3 4 5

Feedback:

- ❑ Presence & Non-Verbal
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

1 2 3 4 5

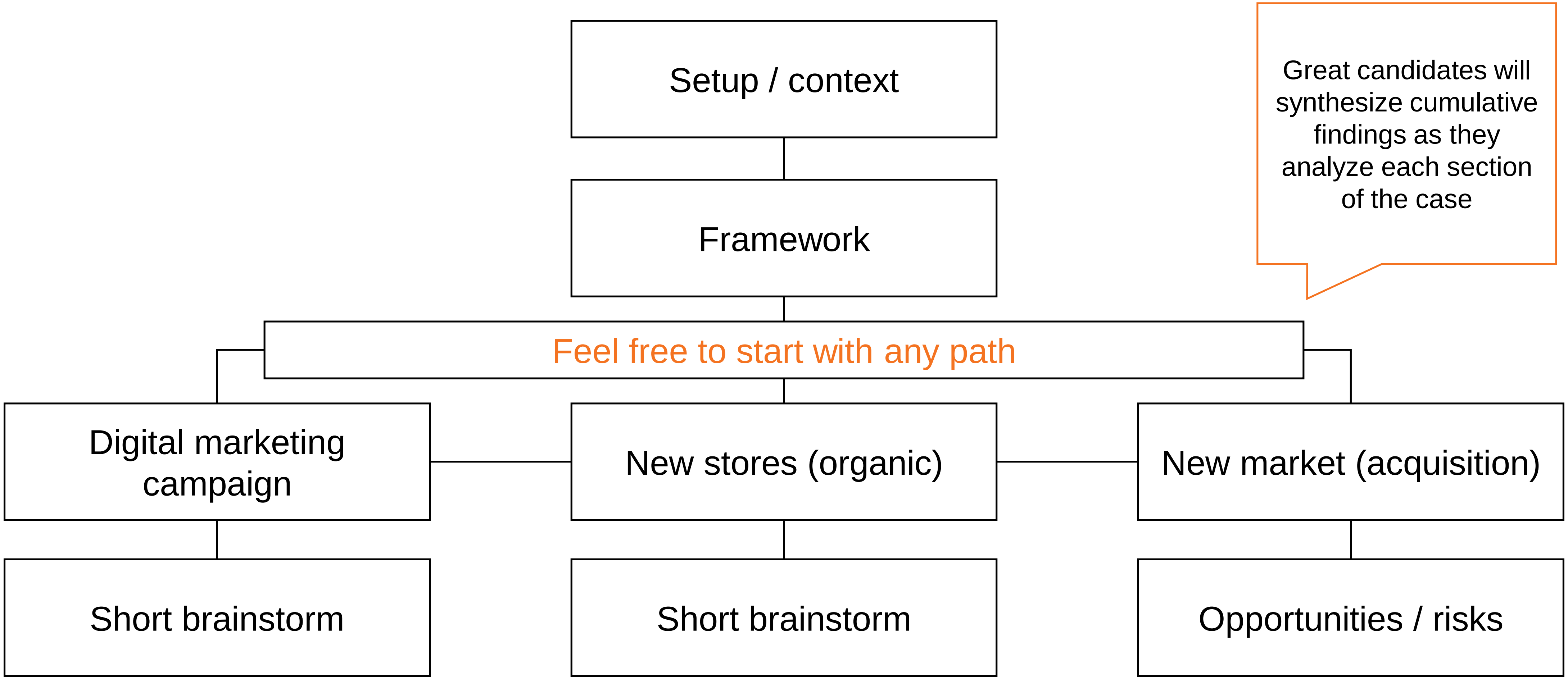
Feedback:

Total: ______ / 20

Smiles4Life

Healthcare

CASE STRUCTURE



SMILES4LIFE

Healthcare

Prompt: Smiles4Life is a regional dental company with 40 locations throughout Texas. The company is a dominate player in both Dallas and Houston, known to have especially strong ties to the Latin community. Last year, Smiles4Life did $60M in revenue with a 20% EBITDA margin after including corporate expenses. At this point, the owner/founder was ready to retire and decided to sell the company to Red Rock Capital, a PE fund based out of LA. Red Rock is excited about the new investment and is now focused on value creation. Red Rock engaged us to help diligence Smiles4Life and, being so pleased with their work, decided to engage us again to develop a growth strategy for its newest portfolio company. How would you advise?

Clarifying Information: *Note: Provide this only if corresponding questions are asked.*

1. Growth target: Red Rock wants to increase EBITDA by 75% within three years of the investment.

2. Business model: Smiles4Life generates revenue much like any typical dental office. Doctors treat patients. Patients then pay either in cash or through a benefit provider (commercial PPO insurance, Medicaid, HMO, etc…).

3. Red Rock Capital: The PE fund focuses on investments across manufacturing, healthcare, and restaurants. It has successfully scaled many multi-site businesses before, especially in the restaurant industry.

BEHAVIORAL INTERVIEW QUESTION:

1. How does consulting fit with your background?

2. When have you gone above and beyond what was required of you?

Framework Guidance:

- There are many possible frameworks to address this case
- Candidates should brainstorm a variety of growth initiatives. Ideas could include:
  - Existing vs new customers, existing vs new services (2x2 could be used here).
  - New products: orthodontics, oral surgeries, implants, whitening, electric toothbrushes, in-house financing
  - Generating top-line demand via AIDA marketing funnel (Awareness, interest, desire, action)
  - Geographic expansion, either organic or acquisition
  - Pricing strategy
  - Cost analysis and reduction: procurement, doctor salary, labor wages, centralization (customer service, claims management)

How to Move Forward:

There is no particular order to how the candidate should run through the case. This is an opportunity for the candidate to guide the conversation. Once the candidate mentions an area of interest that matches with one of the three growth initiatives, move forward to that portion of the analysis.

If the candidate wants to explore other ideas, use any of the following suggested responses:

- Management isn’t interested in exploring that opportunity at this time.
- Management focused on this area recently and doesn’t believe additional opportunities exist right now (for example, cutting costs or adding new products)

Marketing Initiative Management believes there is an opportunity to revamp the marketing efforts of Smiles4Life. This would include a complete overhaul and modernization of the company's website, as well as increasing spend on SEO and other forms of targeted digital ads. **(Hand out exhibit).**

- *Tell candidate:* The variable cost to see one incremental patient is 50% of revenue
- *Share only when asked:* The average spend per dental appointment is $300

Exhibit or Question Guidance:

Provide guidance on how to solve question 1 in the space below.

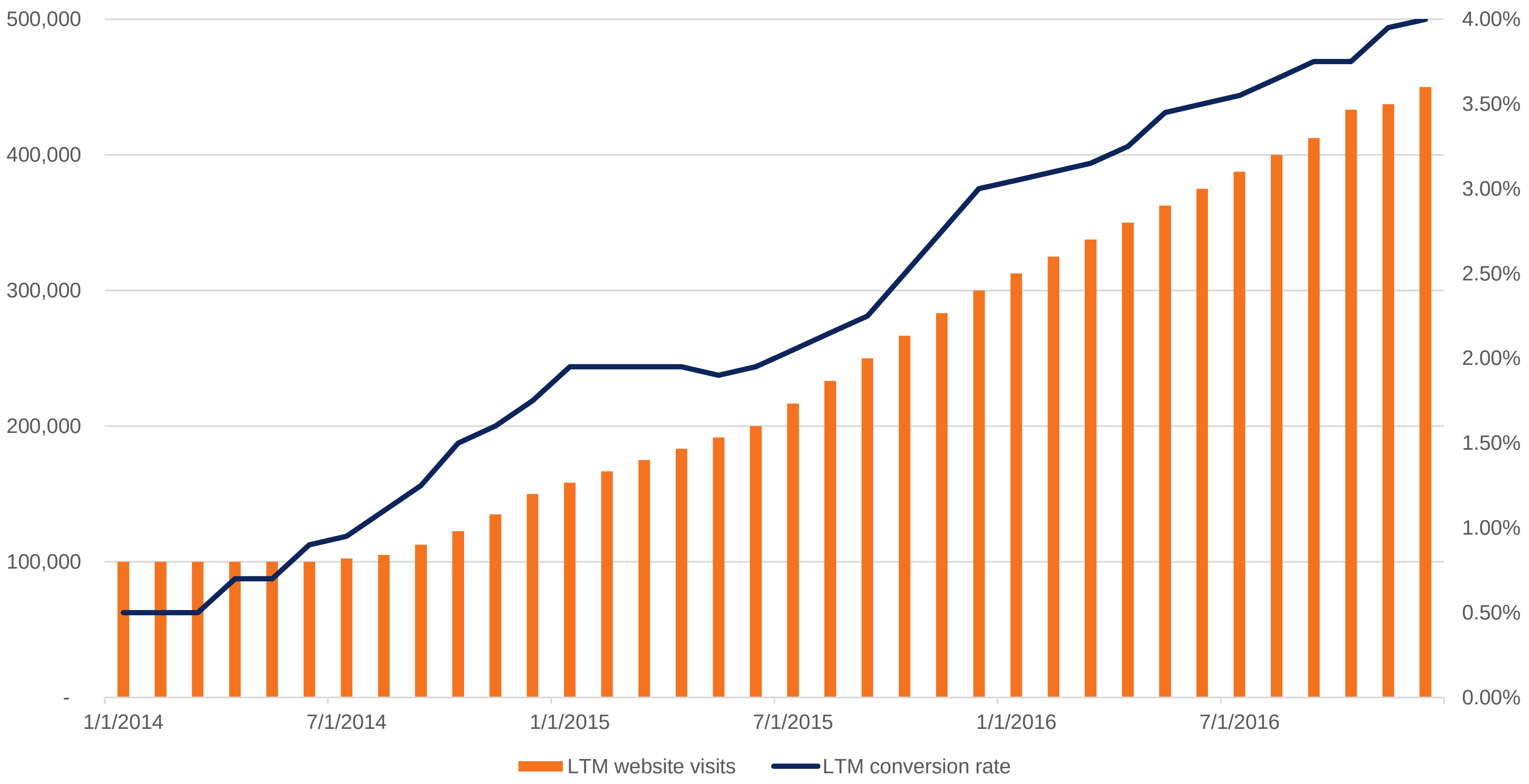
- Calculation explanation:
  - By the end of the third year, LTM (last twelve months) incremental website visits will be 450,000
  - With a 4% conversion rate, this generates 18,000 incremental dental appointments
  - 18,000 appointments x $300 avg spend = $5,400,000 revenue opportunity
  - With 50% variable costs to see an additional patient ($5.4M x 50%) = **$2,700,000 EBITDA opportunity in year three**
- The best candidates put this in context of the EBITDA goal of $9M by year three. This initiative creates **30% of the EBITDA goal.**

What factors should the client consider as it develops and executes this new marketing initiative?

- Customer segmentation / analysis – know who to target and what types of promos they respond best to
- Upfront and on-going cost to update the website
- Marketing ROI by channel (SEO, PPC, Webiste, etc.)

MARKETING

Estimated incremental results of multi-pronged digital program



New stores If the candidate mentions geographic expansion via organic growth, provide the following exhibit. Guide the candidate to calculate the EBITDA opportunity in year 3.

- Note: store EBITDA is higher than company wide EBITDA margin because it does not include corporate expenses

Exhibit or Question Guidance:

- It takes four years for a de novo to reach EBITDA margin maturity
- EBITDA opportunity at the end of year 3 is $4.8M
  - This represents 53% of EBITDA target of $9.0M
- Great candidates will synthesize cumulative findings after evaluating each growth initiative
  - For example, if candidate has evaluated marketing and new stores, they state that together, both initiates create $7.5M of EBITDA, representing a combined 83% of goal

		2014	2015	2016	2017
Tenure at the end of Y3		3	2	1	NA
EBITDA during Y3 / store	$	350	300	200	NA
EBITDA end of Y3 all stores	$	1,400	1,800	1,600	NA
Total EBITDA end of Y3				$ 4,800	

What are the risks and opportunities to consider?

- Opportunities
  - May be easier / less costly than acquisition
  - Strong de novo playbook having put up 40 locations already
  - Opportunity to accelerate pipeline and pull 2017 8+ stores into three-year time horizon?
- Risks
  - Market saturation – do “A” performing locations still exist or will these be suboptimal locations due to saturation
  - Operational burden to expand so quickly (hiring clinical staff, training, and performance management)

DE NOVOS

Average expected de novo performance and pipeline

($ in 000's)		**Year 1**	**Year 2**	**Year 3**	**Year 4**
Office Revenue	$	1,000	1,350	1,400	1,500
Office EBITDA Contribution	$	200	300	350	450
% of Office Revenue		20.0%	22.2%	25.0%	30.0%
		2014	**2015**	**2016**	**2017**
Proposed De Novo Schedule		4	6	8	8+

1. The term de novo is a Latin expression used in English to mean "from the beginning", "anew." Many retailers refer to new, organic stores as "de novos." In short, de novos are new stores.
2. Assume proposed de novos open on January 1st of the respective year

New market via acquisition The management team is interested in expanding into San Antonio. The business development team just identified a potential acquisition – The Smile Center, comprised of 6 dental offices. Each store does about $1MM in sales with a 20% EBITDA margin, representing an estimated $1.2MM EBITDA opportunity. (No need for additional math.)

- What are the opportunities and risks to consider?

Exhibit or Question Guidance:

- Opportunities
  - As the business stands today, this increases EBITDA by $1.2MM or ~13% of EBITDA target
  - Immediate access to cash flow and EBITDA margin
  - Economies of scale when entering a new market (purchasing, regional management)
  - Likely revenue and cost synergies should increase EBITDA dollars and margin
- Risks
  - Deal costs to hit P&L (likely a non-issue for year 3)
  - Integration of people, processes, and culture can be difficult
  - Strategic fit – does the target serve the same customer segments?
  - Comparative economics versus organic market entry strategy

CONCLUSION

We’re about to jump on the phone with the managing director of the PE fund. Please provide a recommendation.

There is no single answer to this case. However, the strongest candidates

- Will complete / evaluate most or all three growth initiatives
- Synthesize cumulative insights after each analysis
- Make a clear, balance, and prioritized recommendation based on the timing, size, and risk of the EBITDA opportunities

Recommendation: A likely conclusion is that the company should:

- Begin execution on digital marketing campaign immediately, especially given long ramp up period
- Expand geographically via organic de novos. Accelerating the pipeline will help ensure EBITDA target is hit in three years
- Further research strategic and financial benefit of The Smile Center acquisition

INTERVIEWER FEEDBACK FORM

Case Name ______________________________ **Interviewer** ________________________________

Case Book ________________________ **Case Type** ______________ **Difficulty** ______________

Case Execution:

- ❑ Clarifying Questions + Framework
  - ❑ Good Questions
  - ❑ Structured
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  - ❑ Creativity

1 2 3 4 5

Feedback:

- ❑ Exhibits + Quantitative Ability
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

1 2 3 4 5

Feedback:

- ❑ Brainstorm + Conclusion
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

1 2 3 4 5

Feedback:

- ❑ Presence & Non-Verbal
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

1 2 3 4 5

Feedback:

Total: ______ / 20

Digging for Gold

Mining

DIGGING FOR GOLD

Mining

Our client is an Australian mining company, whose main product is Gold, which it sells exclusively to China. This company is the largest producer in volume in the Chinese market with 200 tons sold each year. It is also the lowest cost producer at $1000 per ounce of production costs. We estimate the total Chinese demand for Gold today to be around 1500 tons per year. Our client has won a concession to mine a new site adjacent to its biggest mine, and increases production to 300 tons per year (i.e. 100 additional tons per year). Is this worth doing?

Clarifying Information:

Note: Provide this only if corresponding questions are asked.

Are there are any company criteria to approve projects? What typically constitutes success? Board typically approves projects with payback in less than 5 years. You can use payback with no discounting for your math

Geography related question… Mine is adjacent to the original mine and has

How is the market expected to grow? Consider that the market will remain flat at 1500 tons per year for the foreseeable future

How much upfront investment will be required for this project? Prompt to wait until we dive into case ($750M)

What is the cost of this new volume of production? Are there cost synergies or is this a more expensive mine? Prompt to wait until we dive into case (same, $1000/oz)

How does the competitive landscape look like? You can choose to give Exhibit A, but may throw them off receiving before framework

BEHAVIORAL INTERVIEW QUESTION:

1. Why do you want to pursue a career in consulting?

FRAMEWORK GUIDANCE NOTE: *THERE ARE MANY POSSIBLE ALTERNATIVES TO THIS FRAMEWORK. THESE ARE ONLY PROVIDED AS POSSIBLE SUGGESTIONS*

- Undiscounted Cash Flow
  - (plus) Incremental Revenues (volumes increase, but would $/ton go down since it is a commodity
  - (Minus) Direct Operating Costs (candidate should list some possible options)
    - Labor, Repairs & Maintenance, Royalties, Production Taxes, Freight (land & ocean) Processing/Cleaning Costs, Fuel, Utilities
  - +/- incremental indirect (synergies from shared overhead or additional semi-variable costs
  - (minus) Upfront Investment

- Operational Challenges
  - Skilled labor Supply
  - Purchasing
  - Distribution
  - Royalty Negotiation with Australia
  - Safety stock for key machinery

- Macro risk Factors
  - China
    - Trade Relations
    - GDP slow down
  - Supply Shifts
    - Competitor Reaction
    - M&A Activity
  - Shipping Cost Shifts
  - Mining/Environmental Regulatory Changes

How to Move Forward:

To get to the next portion of the case, the interviewee should ask to explore: **Current price/ton and how much that would change with the increased supply**

EXHIBIT 1 – CHINA'S CASH COST CURVE (INCLUDES ROYALTIES AND FREIGHT)

Exhibit Guidance:

The interviewee should notice and verbally acknowledge the following in Exhibit 1. This is where you insert question and the answers to the question

Expected Insights:

1.Increasing capacity pushes out the supply curve lowering the sales price per ton

2. In commodity market, the price is set at the cost of the last ton demanded in the market. (I.e. ton 1,500 cost $1,200). Adding 100 more tons of supply moves that to $1,150

3.We need to understand if the increase volume is offset by lower price and investment cost

How to Move Forward:

They should prompt to calculate payback period of investment. Interviewer should ask for the following

Oz to ton conversion: 15 oz/lb, 200 lb/ton

Investment cost: $600M

<u>Calculate</u>

Before: $1,200 ($/oz) - $1,000 (cost/oz) = $200 profit/oz * 15 oz/lb * 200lb/ton * 200 tons = $1.2B profit/year

After: $1,150 ($/oz) - $1,000 (cost/oz) = $150 profit/oz * 15 oz/lb * 200lb/ton * 300 tons = $1.35B profit/year

Incremental Profit: $1.35B - $1.2B = $150M. $750M/$150M = 5 year payback

BRAINSTORM

Question 2: If you were competitor B, how would your react?

Interviewee should assess that competitor B will lose $50 margin ($1200 - $1150 price drop), and therefore a total of $50 * 200 tons * 15 oz/lb * 2000 lb/ton = $300M per year.

- Interviewee should consider a few options that competitor B has:
- Competitor B can increase his production if they have access to new mines (price will drop even further, but perhaps volume increase will compensate)
- Competitor B can temporarily reduce production to make prices go up again
- Competitor B can work to reduce costs
- Competitor B can assess M&A options (e.g. Higher cost players that are looking to sell, and can potentially have synergies with B's current operations)

A great answer would point out that competitor B needs to expand by another 150M tons (75% increase in capacity before the price would drop lower). Therefore, we are not as concerned about their reaction.

CONCLUSION

You can go either way with the conclusion, but recommend expansion option

Expand

Recommend client expand ~150M incremental profit will payback in 5 years. The larger scale will give us more scale for supplier power and flexibility to bring on/take off supply advantageously.

Concern and Next Steps (just one example of many options)

Shortage of skilled labor force – Partner to build a attraction and retention labor force strategy

INTERVIEWER FEEDBACK FORM

Case Name ______________________ **Interviewer** ______________________

Case Book ________________ **Case Type** __________ **Difficulty** __________

Case Execution:

- ❑ Clarifying Questions + Framework
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

1 2 3 4 5

Feedback:

- ❑ Exhibits + Quantitative Ability
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

1 2 3 4 5

Feedback:

- ❑ Brainstorm + Conclusion
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

1 2 3 4 5

Feedback:

- ❑ Presence & Non-Verbal
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

1 2 3 4 5

Feedback:

Total: _____ / 20

PharmaCo

Healthcare

PHARMACO

Healthcare

Prompt: PharmaCo is a pharmaceutical company with $10 billion in annual revenue. It's corporate HQ and primary R&D centers are in Switzerland, with regional sales offices worldwide. PharmaCo is interested in entering a new, rapidly growing segment of drugs called "biologicals." To gain the R&D capabilities requisite for biologicals, PharmaCo is considering acquiring BioLead, a biologicals start-up in Austin. BioLead is privately owned and has an estimated valuation of $1 billion. Our firm has been hired to evaluate the BioLead acquisition and to advise on its strategic fit with PharmaCo's biologicals strategy. What factors should the team consider when evaluating whether PharmaCo should acquire BioLead?

Clarifying Information:

1. What is PharmaCo's core business? GP has a long, successful tradition in researching, developing, and selling "small molecule" drugs. This class of drugs represents the vast majority of drugs today, including aspirin and most blood-pressure or cholesterol medications.

2. Is entry-by-acquisition the only approach we should consider? R&D for biologicals is vastly different from small-molecule R&D. Since its competitors are already several years ahead of PharmaCo in the biologicals market, PharmaCo wants to jumpstart its biologicals program via acquisition.

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a time you worked with a difficult team member and how you resolved it.

Question 1

- What factors should the team consider when evaluating whether PharmaCo should acquire BioLead?

Exhibit or Question Guidance:

A **good answer** would include the following:

- The value of BioLead’s **drug pipeline**, number of drugs currently in development, likelihood of success, estimated revenues and profits
- BioLead’s **R&D capabilities** (future drug pipeline), scientific talent, intellectual property (for example, patents, proprietary processes or know-how for biologicals research), and buildings, equipment, and other items that allow BioLead’s R&D to operate.
- BioLead’s **marketing or sales capabilities**. Especially how promotional messages will be delivered, for example, relationships with key opinion leaders that can promote biologicals; key opinion leaders can come from the academic arena, like prominent medical school professors, or from the public arena, like heads of regulatory bodies or prominent telejournalists.
- **Acquisition price**

A **very good answer** might also include:

- BioLead’s existing partnerships or other relationships with pharmaceutical companies.
- PharmaCo’s **capability gaps** in biologicals, R&D, sales and marketing, etc.
- PharmaCo’s **alternatives to this acquisition**. Alternative companies PharmaCo could acquire. Other strategies for entering biological segment, for example, entering partnerships rather than acquiring, and pursuing other strategies than entering the biological segment

BRAINSTORMING

The team wants to explore the value of BioLead's current drug pipeline. Based on market research, BioLead's only promising drug, SM1, is estimated to generate $10B in sales if brought to market. That said, **what costs should be considered throughout the entirety of a drug's lifetime?**

Brainstorming Guidance:

Note: This is just one possible set of categories and answers. Many more are possible, and interviewers should push candidates to be specific and comprehensive.

Research

- Highly skilled labor (scientists)
- Specialized equipment and labs
- Materials
- Phase I, II, III testing – proper testing and documentation
- Cost of failed drugs

Regulatory approval

- Regulatory fees paid to governing bodies
- Salary / wages for in-house council to file patents, trademarks, and other legal documentation

Commercialization

- Promotional materials
- Marketing campaigns
- Production costs (materials, quality, sourcing)
- Adding personnel (sales, marketing, administrative, regulatory)
- New facilities
- Packaging materials
- Distribution and shipment (logistics)
- Taxes
- Patent infringement

Best candidates display:

The best candidates will apply a customized structure and will brainstorm cost items specific to pharmaceuticals. Push candidate to produce a comprehensive list by asking "What else?" one or two times.

Question 3

- The team has pulled together the following data (show Exhibit 1). What is the value of BioLead's drug, SM1?

Supply the following information only when asked:

- Costs by phase: Phase I $160M, Phase II $125M, Phase III $75M, Filing $5M. Costs are incurred only if the drug reaches a particular phase.
- Costs of production: Manufacturing 5% of sales, Logistics 5% of sales, Other 10% of sales
- Lifetime revenue: $10B – **great candidates** will recall that this information was previously provided

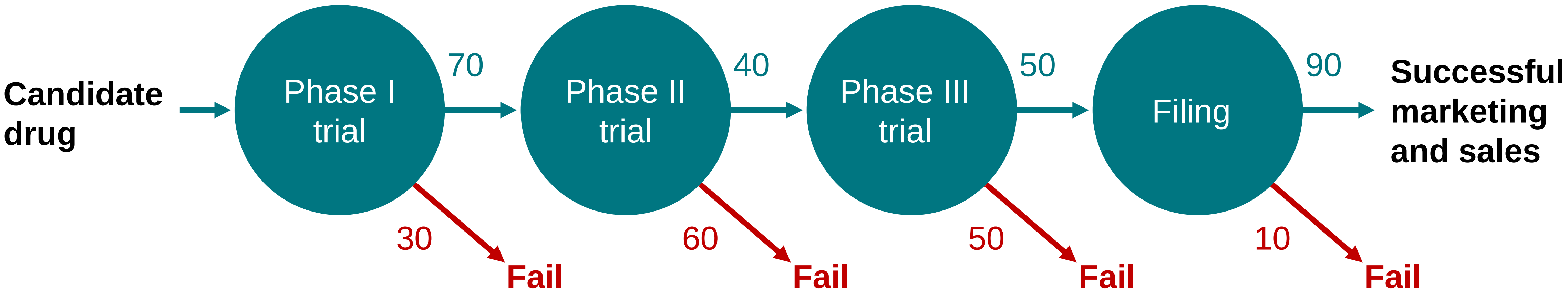
Question Guidance:

- Revenue (discounted by chance of success): $10B x 70% x 40% x 50% x 90% = $1,260MM
- Production costs: $10B x (5% + 5% + 10%) = $2B, then discounted by chance of success = $252MM
- Phase 1 has a 100% chance of completion = $160MM
- Phase 2 achieved 70% of the time (probability a drug completes Phase 1) = $125M x 70% = $87.5M
- Phase 3 achieved 70% x 40% of the time = $75M x 70% x 40% = $21M
- Filing achieved 70% x 40% x 50% of the time = $5M x 70% x 40% x 50% = $0.7M
- Revenue ($1,260MM) – Production costs ($252MM) – R&D costs ($269.2MM) = **$738.8MM SM1 drug valuation**

This drug valuation essentially represents known future cash flows of the business. **Great candidates** will compare this figure against the $1B BioLead company valuation mentioned in the prompt and hypothesis why the figures are different.

EXHIBIT 1

Expected probability of success, by stage of research and development, %



Note: "Filing" is the process of submitting all of the clinical and safety evidence from Phase I, II, and III trials, and asking for regulatory approval to actually sell the drug.

Question 3

- What are your hypotheses on the major risks of integrating the R&D functions of BioLead and PharmaCo?

A very good answer would include the following:

- Little to no overlap in research or expertise leading to minimal collaboration
- Culture clash. PharmaCo is an established, mature business while BioLead is a young, entrepreneurial business
- Language barriers hinder communication and sharing of information
- Physical distance and time difference may lead to a poor sense of community
- Talent may leave BioLead after the acquisition – either as a result of newfound wealth from the sale of the business or because they don't want to be a part of a large corporation

The **best candidates** will recognize the human element of organizational change

INTERVIEWER FEEDBACK FORM

Case Name ______________________________ **Interviewer** ________________________________

Case Book ________________________ **Case Type** ______________ **Difficulty** ______________

Case Execution:

- ❑ Clarifying Questions + Framework
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

1 2 3 4 5

Feedback:

- ❑ Exhibits + Quantitative Ability
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

1 2 3 4 5

Feedback:

- ❑ Brainstorm + Conclusion
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

1 2 3 4 5

Feedback:

- ❑ Presence & Non-Verbal
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

1 2 3 4 5

Feedback:

Total: ______ / 20

Shisha: just blowing smoke?

Market Entry

SHISHA: JUST BLOWING SMOKE? Market Entry

Prompt:

Over the past five years, the government of Saudi Arabia has been focused on reducing economic dependence on oil by diversifying the domestic economy. As part of this effort, the government has evidenced a willingness to relax certain social norms. As part of the diversification initiative, the government hired us to forecast the potential revenue impact of taxing shisha consumption.

Clarifying Information: *Note: Provide this only if corresponding questions are asked.*

1. **What is shisha?** Shisha is an instrument for vaporizing and smoking flavored tobacco. In the Arab world and Middle East, people smoke waterpipes as part of the cultural traditions.
2. **How will the Saudi government make money off of shisha?** Institutions must apply and pay for an annual license to sell shisha. Additionally, sales tax and import tariffs are levied by the government as appropriate.
3. **Is there a specific revenue target in mind?** No, the Saudi government only wishes to maximize revenue over a 3-year period.
4. **How much revenue does the Saudi government currently earn?** 2.5 trillion Saudi riyal annually. Tobacco and other related products currently generate 500 million SAR in revenue.
5. **Are there any social norms we should be aware of?** Islam is widely practiced by Saudis citizens and governs their personal, political, economic and legal lives. Additionally, public consumption of shisha is currently banned but is widely consumed at home.

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a time you had an ethical dilemma.

Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions.

Revenue Sources

- Licenses
  - Type and quantity of businesses that will sell shisha
  - Application payment process for licensing: one-time fee vs. annual fees
  - Lifetime of typical business selling shisha
  - Adoption rate over three years
- Sales Tax
  - Consumer segments and spending habits
  - Adoption rate of public use
  - Sales tax rate
- Import Tariffs
  - Quantity of domestic vs foreign sourced shisha consumption
  - Tariff percentage

Cost Considerations

- Regulatory body
- Auditing and quality control
- Implementation vs on-going costs

Social Issues

- Public opinion of legalizing public consumption of shisha
- Health & well-being of citizens
- Job creation

How to Move Forward:

The candidate should include criteria to size the revenue opportunity over a three-year period and should want to naturally start exploring this aspect. If the candidate wants to explore something else first, ask why? Then point him/her in the direction of sizing revenue.

Note: Inform the candidate that the client is mainly interested in estimating revenue from issues licenses. Hand interviewee Exhibits 1 & 2.

EXHIBIT 1

Shisha licensing structures

Country	License Structure
Qatar	10,000 Saudi riyal one-time fee
Jordan	3,000 Saudi riyal one-time fee + 500 riyal monthly fee
United Arab Emirates (UAE)	6,000 Saudi riyal one-time fee + 3,000 Saudi riyal annual fee

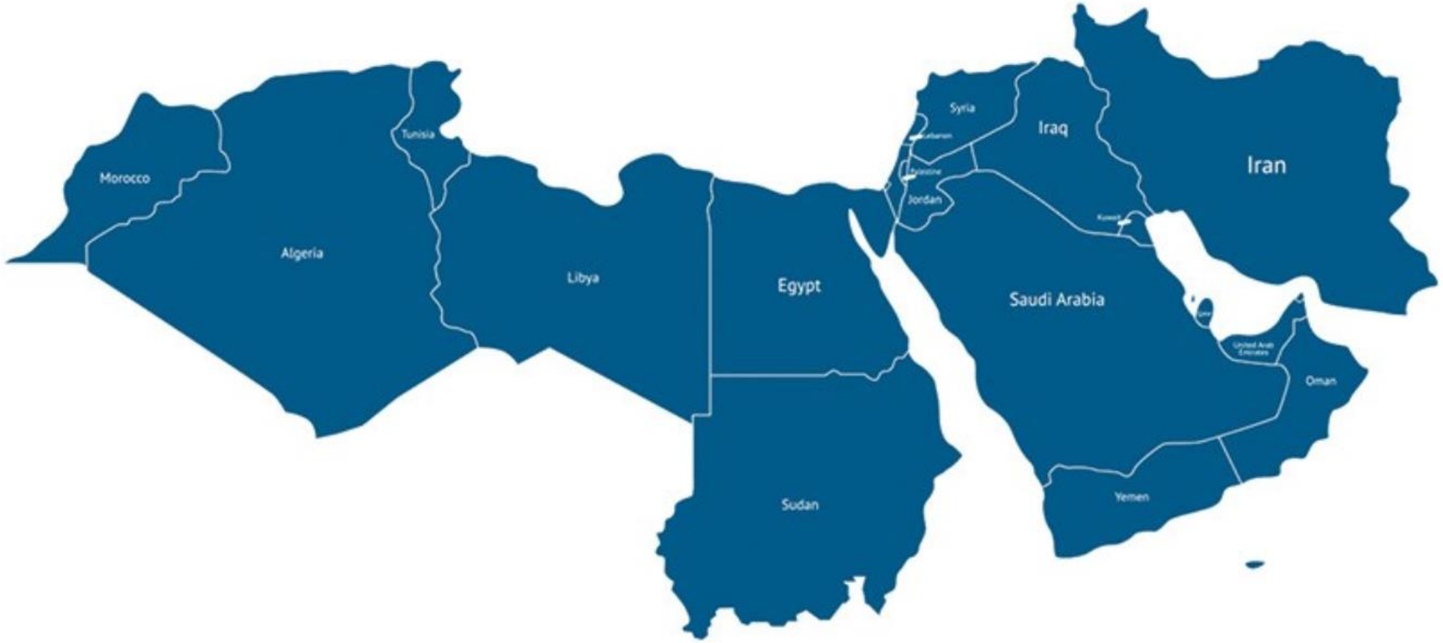


EXHIBIT 2

Riyadh (capital of Saudi Arabia), restaurant market segmentation

Restaurant Type	Fast Casual	Casual Sit-Down	Fine Dining	Tea & Coffee Shops
Number of Businesses	5,000	2,000	1,000	6,000
U.S. market equivalents	Chipotle, Cava, McDonalds	Panera, Applebee’s	C&O, Ivy Inn Restaurant	Starbucks

*The capital city of Riyadh represents about 20% of the domestic restaurant market

*The average lifespan of a restaurant is three years

Question 1

- Which licensing structure should Saudi Arabia adopt? What is the estimated three-year revenue impact?

Supply only when asked: *Because of unusual economic growth, 1,000 new restaurants will be added to the Riyadh region each year for the next two years.*

Supply only when asked: *Related products have experienced an adoption rate of 5%, 10%, and 15% in years 1, 2, 3 respectively.*

Exhibit or Question Guidance:

After looking at Exhibits 1 & 2, candidates should be able to identify and call out the following:

- To maximize and forecast revenue, the candidate must execute two analyses: (1) identify which licensure structure to pursue and (2) apply license structure to restaurant industry to estimate four-year revenue impact
- Assume shisha licenses will only be sold to restaurants
- Jordan's license structure maximizes license revenue

	One-Time Fee	Annual Fee			
	Year 1	Year 1	Year 2	Year 3	Total
Qatar	10,000				10,000
Jordan	3,000	6,000	6,000	6,000	21,000
UAE	6,000	3,000	3,000	3,000	15,000

Exhibit or Question Guidance Cont:

Once the Jordan license structure has been identified, the candidate should calculate year 3 revenue:

- Calculate the number of total and newly participating businesses. This will enable you to calculate the one time fee in year 3.
- Annual fees (500*12) generate 6k riyal in revenue per participating business.
- Calculate total domestic revenue assuming Riyadh represents about 20% (i.e. multiply by 5).
- Great candidates will put the 85.5 million riyal into perspective
  - Current revenue related to Tobacco related products is 500 million riyal
  - This initiative will increase revenue by 17.1% in three years

	Year 1	Year 2	Year 3
Riyadh Businesses	14,000	15,000	16,000
Adoption Rate	5%	10%	15%
Total Participating Restuarants	700	1,500	2,400
Newly Participating Restaurants	700	800	900
In Millions Riyal			
One Time Fee (3k)	2.10	2.40	2.70
Annual Fee (6k)	4.20	9.00	14.40
Total License Revenue	6.30	11.40	17.10
Riyadh Percentage of GDP			20%
Estimated Domestic Revenue Impact (millions riyal)			85.5

BRAINSTORMING

Are there any other considerations the Saudi government should keep in mind before legalizing the public consumption of shisha?

Brainstorming Guidance:

This is a “what else” section. Below are some basics but ideally, you’re looking for the interview to be as creative as possible. As with most questions of this type, a bad answer will stop at one or two. A good answer will have a creative list. A great answer will have a structure that makes the answer MECE. A great answer should also prioritize the findings indicating which ones he/she thinks are the most important.

Key Points to Consider:

- Health and well-being of citizens given increased consumption of tobacco products
- Other distribution channels to sell shisha other than restaurants
- Associated costs with launching and governing program
- Public opinion – restaurants may not want to sell shisha and patrons may not want to smoke in public
- Reputation on the global stage. Many countries are restricting use of tobacco products
- Revenues related to sales tax and tariffs
- Job creation related to budding industry
- Three-year economic forecast

CONCLUSION

The minister of commerce is dialing in on Skype and expects a summary. Please share your findings.

Recommendation:

- Candidate should succinctly summarize findings, including the license structure that optimizes revenue and year 3 revenue opportunity.
- Great candidates will note that calculated figures are conservative as they do not include revenue from sales tax or import tariffs.

Risks:

- License revenue depends heavily on adoption rate
- Healthcare costs related to increased tobacco use

Next Steps:

- Market analysis of restaurant's willingness to sell shisha (i.e. adoption rate)
- Explore the financial and social costs related to the program

INTERVIEWER FEEDBACK FORM

Case Name ______________________ **Interviewer** ______________________

Case Book __________________ **Case Type** ___________ **Difficulty** ___________

Case Execution:

- ❑ Clarifying Questions + Framework
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

1 2 3 4 5

Feedback:

- ❑ Exhibits + Quantitative Ability
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

1 2 3 4 5

Feedback:

- ❑ Brainstorm + Conclusion
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

1 2 3 4 5

Feedback:

- ❑ Presence & Non-Verbal
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

1 2 3 4 5

Feedback:

Total: _____ / 20

MBS Co.

Financial Institutions

MBS CO.

Financial Institutions

Prompt:

Your client is MBS Co., a government sponsored enterprise (GSE), that purchases mortgages from banks, packages them into mortgage-backed securities (MBS), and sells them in the secondary market to investors. MBS Co.'s return on capital (ROC) has declined since last year, and the management team is looking for your help in understanding why this has happened and how to improve ROC going forward.

Clarifying Information: *Note: Provide this only if corresponding questions are asked.*

- MBS Co. buys individual mortgages from banks and packages several thousand of them into a given security. Shares in that security are sold to investors.
- When a homeowner makes a monthly mortgages payment, the payment passes through the bank first, then through MBS Co., then to investors (primarily institutional investors).
- MBS Co. keeps a small fee on each payment that passes through. The fee amount is based on a percentage of mortgage volume (i.e. how much money was loaned to homeowners).
- MBS Co. has one other GSE competitor that buys mortgages and creates mortgage-backed securities. The securities from both companies are fungible to investors.
- Banks can also choose to hold mortgages – rather than sell to a GSE – and keep the full monthly mortgage payments from homeowners.
- There is no target ROC. Any improvement in ROC would be considered a success.
- Return on Capital (ROC) = profits / capital

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a time you failed.

Framework Guidance:

- Framework should identify profit and capital as components of ROC (traditional profit tree w/o capital is not sufficient)
  - MBS Co.'s fee as a percent of mortgage volume has been consistent since last year, and the fee percent is consistent for all mortgages (regardless of riskiness)
  - Internal costs (e.g. SG&A) are negligible
  - Regulations have not changed
- A good framework will call out competition (other buyers of mortgages) and/or vertical integration in the value chain
  - No new competitors have entered the market; the government only allows MBS Co. and its one competitor to create MBS
  - Banks have begun holding more mortgages (if candidate asks about this, indicate it will be discussed later in the case)
- A good interviewee will note market conditions as a potential reason for the decrease in ROC but will recognize this is not a lever MBS Co. can use to increase ROC going forward
  - All market conditions have remained consistent since last year

Example framework

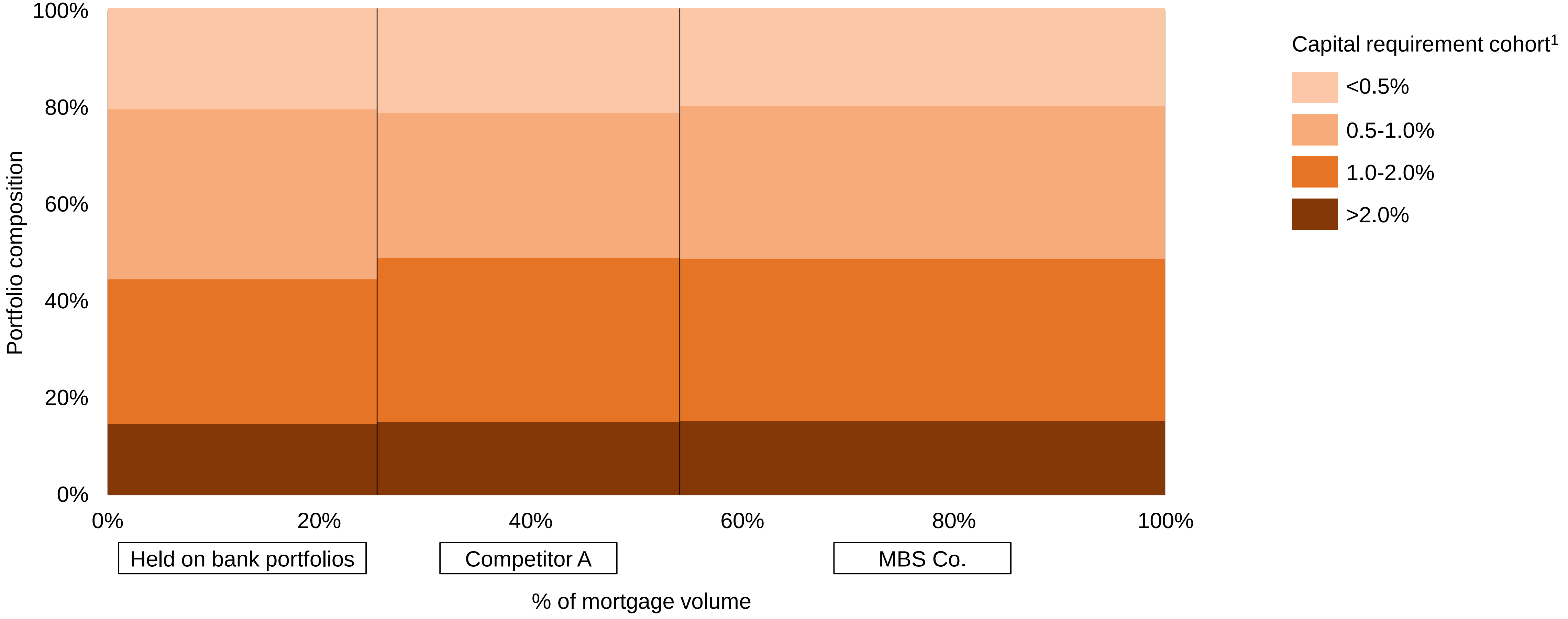
ROC	Profit	• Mortgage riskiness (borrower credit worthiness) • Pricing pressure from banks • Fee pressure from investors • Internal costs
	Capital	• Regulatory requirements • Excess cash build-up • Build-up of unsecuritizable mortgages (i.e. bought from bank but can't turn into MBS)
Market		• Interest rates, yield curve • Oversupply of housing • Consumer housing preferences
Competition		• New entrant • Competitor pricing/fees • Customer/supplier vertical integration (e.g. banks hold mortgages)

How to Move Forward:

- The interviewee should identify capital held as a key lever in the ROC calculation and inquire as to whether it has changed recently
  - If the interviewee needs coaching ask them what they think the components of ROC are
- Provide the interviewee with Exhibits 1 & 2

EXHIBIT 1

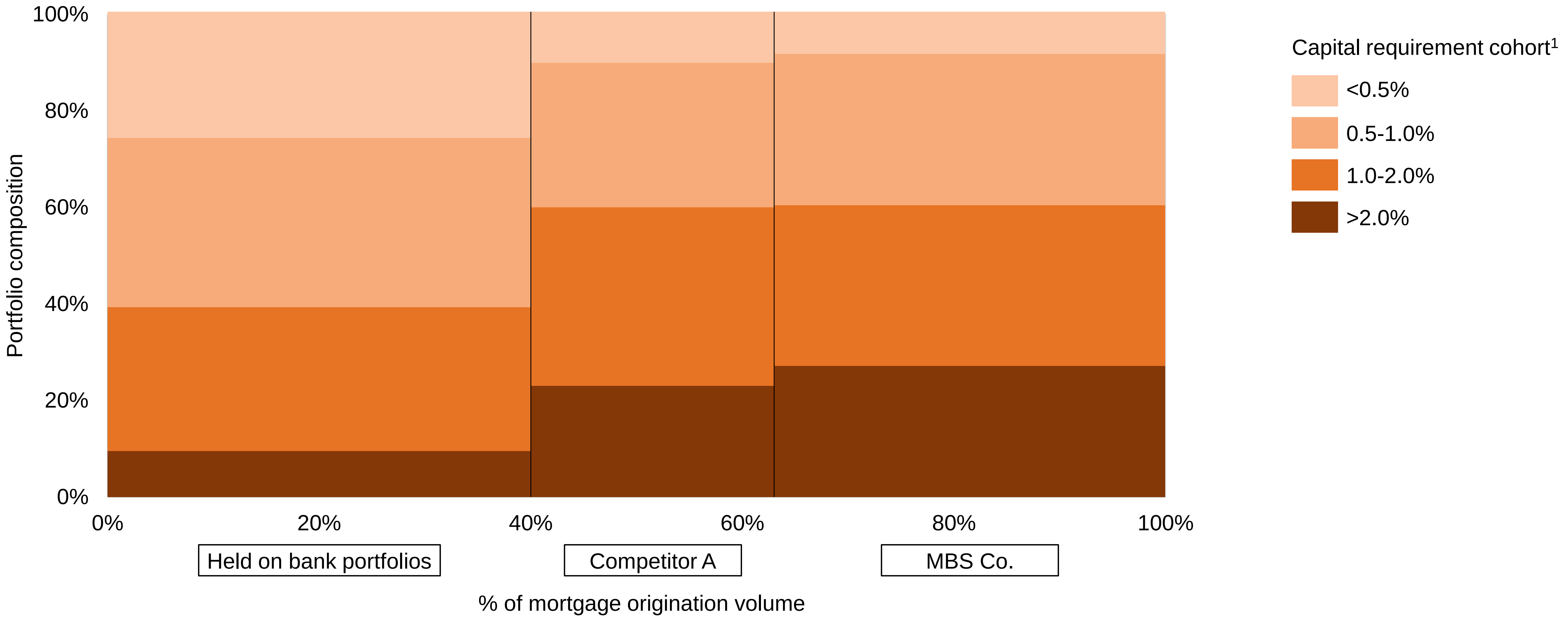
2017 mortgage originations



[1]Capital required to be held per dollar of mortgage securitization volume. Based on FICO score and loan-to-value ratio of borrower/mortgage.

EXHIBIT 2

2018 mortgage originations



[1]Capital required to be held per dollar of mortgage securitization volume. Based on FICO score and loan-to-value ratio of borrower/mortgage.

Exhibits 1 & 2 Guidance:

- Exhibit description
  - X-axis reflects market share as measured by mortgage volume securitized (MBS Co. and Competitor A) or held on portfolio (banks)
  - Y-axis reflects breakdown in portfolio of each company by cohort
  - Capital requirement represents how much cash (as a % of mortgage volume) a company must hold on its balance sheet to support a given mortgage
- Points of clarification (provide if interviewee asks)
  - Total market size in dollars has remained consistent
  - Cohort ranges correspond with capital requirements as specified by the government
  - Total market composition of mortgages by cohort has remained consistent (i.e. same number of high/low quality mortgages in the market; any shift in one company's portfolio composition is zero-sum for competitors)
  - Each company's percent of mortgages in the '0.5-1.0%' and '1.0-2.0%' cohorts is consistent between years
- Observations (a good interviewee will recognize these)
  - Cohorts with higher capital requirements will generate lower ROC
  - Portfolio compositions were roughly the same for MBS Co., Competitor A, and banks in 2017
  - MBS Co. and Competitor A lost market share to banks holding loans on their portfolios in 2018
  - MBS Co.'s portfolio composition shifted towards high capital requirement mortgages (as did Competitor A's)
  - Portfolio composition of mortgages held by banks shifted towards low capital requirement mortgages
- Key insight (push interviewee to explain why the observations above might be related)
  - Because banks are earlier in the value chain, they can choose to keep good mortgages (low capital requirements) on portfolio and pass bad mortgages (high capital requirements) to MBS Co. and Competitor A
- To advance, interviewee should identify MBS Co.'s portfolio mix shifted to higher capital requirement loans in 2018, while banks are retaining more low capital requirement loans

Exhibit 3

- The client has identified a way for 10% of the capital backing high capital requirement mortgages (>2% cohort) to be redeployed in low capital requirement loans (<0.5% cohort). How much will this shift change overall ROC?
- Provide the interviewee with Exhibit 3

Exhibit Guidance:

- Interviewer notes
  - The interviewee must shift the capital between cohorts and calculate the new wtd avg ROC % for the portfolio
  - Securitization volume is not needed to perform the calculation; its inclusion in the exhibit is intended as a red herring
- Points of clarification (provide if interviewee asks)
  - Capital and securitization volume are related via the capital requirement (e.g. in the '<0.5%' cohort, $500B securitization volume x 0.4% wtd avg capital requirement = $2B capital)
  - Cohort ROC (i.e. ROC within a cohort) does not change as a result of the capital shift; cohort ROC is based on the riskiness of the mortgages that fall in the capital requirement domains
- Takeaway / insights
  - A good interviewee will recognize the shift in capital will drive a ROC increase, before even performing any calculations
  - After completing calculation, interviewee should declare capital shift results in increased ROC, thus satisfying client's goal
  - Second order insight: Shifting capital to lower capital requirement loans will require increase in total portfolio size (and thus, market share)
- To advance: A good interviewee will begin brainstorming ways for MBS Co. to make this shift in its portfolio and/or risks associated with the shift. If they don't proactively do this, provide them with the next question.

EXHIBIT 3

2019F securitization portfolio

Capital requirement cohort	Securitization volume ($, billions)	Capital ($, billions)	Cohort ROC
<0.5%	500	2.0	25.0%
0.5-1.0%	1,200	8.0	15.0%
1.0-2.0%	500	5.0	10.0%
>2.0%	200	5.0	4.0%
Total	2,400	20.0	12.0%

EXHIBIT 3 (MATH DETAIL)

Method 1 (fastest)

- Interviewee calculates change in portfolio ROC % by only examining incremental effects of capital shift

Capital requirement cohort	Change in capital ($B)		Cohort ROC	Change in dollar returns ($M)
<0.5%	+0.5	x	25.0%	+125
>2.0%	-0.5	x	4.0%	-20
Net change	0.0			105

$105m change in dollar returns / $20B in capital = ~0.5% change in ROC

New ROC = 12.5%

Alternate approach: (25.0% - 4.0%) x $500M = $105M

Method 2

- Interviewee calculates change in portfolio ROC % by recreating a full wtd avg calculation for the new portfolio

Capital requirement cohort	Capital ($B)		Cohort ROC	Dollar returns ($M)
<0.5%	2.5	x	25.0%	625
0.5-1.0%	8.0	x	15.0%	1,200
1.0-2.0%	5.0	x	10.0%	500
>2.0%	4.5	x	4.0%	180
Total	24.0			~2,500

$2.5B dollar returns / $20B in capital = 12.5% ROC (new)

Question 1

- What methods can MBS Co. use to shift its portfolio to higher ROC loans?

Question Guidance:

- Identifying options based on the various stakeholders in the mortgage value chain is one effective way of brainstorming this question
- Interviewees may also choose to use an internal/external framework, in which all the non-MBS Co. stakeholders would align with the external bucket

Stakeholders / value chain	
Homeowners	• Build preference for MBS Co. with high quality borrowers by simplifying mortgage process • Improve screening/selection process to better identify high quality borrowers
Banks	• Build preference to sell to MBS Co. by simplifying mortgage process • Lower fees to entice banks to sell high quality mortgages rather than hold on to them • Form partnerships with banks to bolster flow of high quality mortgages to securitization
MBS Co.	• Bypass banks and begin originating mortgages directly with consumers • Expand into other geographic regions (i.e. non-US) • Expand into other types of loans (e.g. student loans, car loans)
Government	• Lobby government to lower capital requirements • Request government capital infusion to fund more high ROC mortgage purchases

CONCLUSION

To conclude, the interviewee should provide the following:

Recommendation:

- MBS Co.'s return on capital has declined because banks are choosing to hold high-ROC mortgages on their portfolio (gaining market share) and passing low-ROC loans to MBS Co. (shift in portfolio). MBS Co. has an opportunity to increase it's return on capital by 0.5% to 12.5% by shifting capital in its portfolio to higher quality loans by… [interviewee should choose an idea mentioned in brainstorming]

Risks:

- [Risks should be tailored to the ideas chosen]
- Potential risks include: upset stakeholders affected by idea chosen, MBS Co.'s lack of capabilities to execute idea

INTERVIEWER FEEDBACK FORM

Case Name ______________________ **Interviewer** ________________________

Case Book __________________ **Case Type** ___________ **Difficulty** ___________

Case Execution:

- ❑ Clarifying Questions + Framework
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

1 2 3 4 5

Feedback:

- ❑ Exhibits + Quantitative Ability
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

1 2 3 4 5

Feedback:

- ❑ Brainstorm + Conclusion
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

1 2 3 4 5

Feedback:

- ❑ Presence & Non-Verbal
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

1 2 3 4 5

Feedback:

Total: _____ / 20

Emerald City

Public Sector

EMERALD CITY

Public Sector

Prompt:

Your client is the mayor of Emerald City - a large, metropolitan US city (i.e. New York or Los Angeles). She has noticed that Emerald City's FDI levels– foreign direct investment – have been decreasing over the past year. The mayor wants to turn this trend around. What are the main factors she should consider as causes of the decline?

Clarifying Information: *Note: Provide this only if corresponding questions are asked.*

1. What is FDI? Foreign direct investment is an investment made by a firm or individual in a foreign country into businesses located in the country of interest. Generally, FDI takes place when an investor establishes foreign business operations or acquires foreign business assets (as opposed to equity purchases).

2. Why does the mayor want to increase FDI? She is interested in the foreign income opportunities, as well as the benefits to Emerald City's population.

3. What foreign entities currently invest in Emerald City? – Foreign investors into Emerald City include firms from a broad range of markets, from mature economies such as the EU and China, to emerging economies such as Brazil and India.

4. What is the goal/target? – A $60 billion increase in FDI over the next 2 years.

5. [Any other question – e.g. "what are Emerald City's major industries"] – Whatever you want them to be/we don't have information on that.

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a conflict you experienced that wasn't fully resolved.

Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions

A good framework will consider this question from multiple angles, including both financial and non-financial elements. An excellent framework will incorporate potential solutions or risks as a secondary focus, but should be a brief mention since the prompt specifically asks for causes. Certain characteristics could be argued for one bucket or another (i.e. political stability could potentially be in Emerald City if referring to local economy and not U.S. or other countries.

	Emerald City Characteristics (Internal)	Macroeconomic (External 1)	Geopolitical/Social (External 2)

Causes	• Growth opportunities
	  ◦ Industry specific factors
	  ◦ Population growth
	• Tax Rate
	• Transport/Infrastructure
	• Labor
	  ◦ Skill of labor force
	  ◦ Wage Rates
	  ◦ Unemployment
		• **Mix/growth of foreign investors**
		• Strength of dollar
		• Diversification potential
			• Free Trade/Tariffs
			• Political Stability/Corruption
			• Geographical constraints

Risks	• Domestic political shift • Integration of production/supplier sites • Cannibalization of domestic investment	• Effect on exchange rates	• Environmental concerns • Cultural tensions

How to Move Forward:

To move forward, the interviewee should call out that they are interested in knowing more about where FDI is currently coming from and the direction in which it is going (or they should mention it in their framework) and that once we understand the magnitude and change of each factor then we can explore ways to improve.

Mention: The mayor has some numbers on a subset of current FDI levels and trends. *Hand the interviewee Exhibit 1.*

EXHIBIT 1

Emerald City FDI sources

Market	GDP (trillions)	FDI in Emerald City*	Growth Rate
EU	19	0.5%	2%
China	12	1%	4%
Canada	1.2	10%	5%
Russia	1.5	2%	10%
Brazil	2	5%	7%
India	2.6	3%	5%

* Expressed as %GDP

Question 1

- The mayor wants to know what you think of her goal to increase FDI.

Exhibit or Question Guidance:

After looking at Exhibit 1, candidates should be able to immediately call out the following:

- Emerging markets have generally higher growth potential than mature ones
  - This is expected due to their size and the law of diminishing marginal returns
- Other callouts, such as Canada's large investment and Russia's large growth rate

Market	Current FDI	Year 1	Year 2	Total Growth
EU	95 billion	~97 billion	~99 billion	~4 billion
China	120 billion	~125 billion	~130 billion	~10 billion
Canada	120 billion	126 billion	~132 billion	~12 billion
Russia	30 billion	~33 billion	~36 billion	~6 billion
Brazil	100 billion	107 billion	~114 billion	~14 billion
India	78 billion	~82 billion	~86 billion	~8 billion

Calculations:

- The ideal candidate does not get too bogged down in exact calculations or flustered by them. In general, you should **let the candidate round and approximate if they ask** where it is reasonable, particularly because this is the mayor asking for a gut check.
- Refer to table for numbers. The difference between current FDI and Year 2 gives total growth. The summation is approximately $54B
- Growth Formula:
  - Easy way: GDP x FDI% x 2 years x Growth Rate
  - Hard way: GDP x FDI% x $(1 + \text{Growth Rate})^2$ *see table above-right for help if candidate attempts this

Takeaway: The candidate should determine that the $60B goal is **not aggressive enough**. Between these markets alone, the current expected growth gets the mayor **90%** of the way there. She can set a higher target if she implements new growth initiatives and includes other foreign market potential.

Once the candidate identifies that the goal should be set higher, move on to Question 2.

BRAINSTORMING (QUESTION 2)

The mayor is interested in your overall thoughts on what increased FDI will do for her and Emerald City, as well as the risks.

Brainstorming Guidance:

Note: This is just one possible set of categories and answers. Many more are possible, and interviewers should assess both the volume and relevance of answers.

Benefits:

- Financial
  - Increased investment/consumption
  - Decreased unemployment
- Non-financial
  - Stability
  - Re-election

Risks:

- Regulatory
  - US Legislation not aligned with local goals
- Feasibility
  - Transportation capabilities/logistics
  - Integration of cultures

Best candidates display:

The interviewee should hit on both benefits and risks, including a mix of financial and non-financial factors, particularly if they did not do so satisfactorily in their framework. A very strong response will also provide mitigating factors for the risks.

Question 3

- Now I am the mayor. Tell me why FDI benefits my citizens.

Exhibit or Question Guidance:

*Your motivation as the mayor is **only** to do what benefits **the well-being** of your citizens.*

This question is in lieu of a standard recommendation conclusion. It is intended to stretch the candidate a bit.

- If the candidate mentions financial or political considerations, redirect them to address **only** the well-being of your citizens.
- Potential benefits:
  - Technology
  - Education
  - Fulfillment
  - Lifestyle (disposable income)
- If the candidate does not answer this question succinctly (~60 seconds)…
  - **Ask:** Would you say all of that to the mayor?
  - The mayor wants a quick answer. Have them try again.

INTERVIEWER FEEDBACK FORM

Case Name ______________________________ **Interviewer** ________________________________

Case Book ________________________ **Case Type** ______________ **Difficulty** ______________

Case Execution:

- ❑ Clarifying Questions + Framework — 1 2 3 4 5
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

Feedback:

- ❑ Exhibits + Quantitative Ability — 1 2 3 4 5
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

Feedback:

- ❑ Brainstorm + Conclusion — 1 2 3 4 5
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

Feedback:

- ❑ Presence & Non-Verbal — 1 2 3 4 5
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

Feedback:

Total: ______ / 20

Omega Naval Systems

Aerospace & Defense

OMEGA NAVAL SYSTEMS

Aerospace and Defense

Omega is an aerospace defense contractor with three naval radar systems in its product lineup.

These systems are equipped on naval destroyers and aircraft carriers.

They have three radar systems in their product lineup, Products 1, 2 and 3. 1 and 3 are substitutes for one another.

The company has experienced declining profits in recent years and has brought us in to advise how to turnaround their financials.

Clarifying Information:

Only sell to US Navy with no intention of expanding to other branches.

Omega makes money on the sale of the radar system, no ongoing service contracts. One time sale.

No new products are expected to be developed or launched.

No target improvement goal, just better than what it currently is.

No historic profitability figures.

All options on the table regarding the 3 product lines.

Expect 100% conversion of customers from products 1 to 3 or 3 to 1 if either is shut down.

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a time you utilized data to influence someone.

RECOMMENDED FRAMEWORK

- Candidate should recognize this is a profitability problem and should drive the case here first. Strong candidates will mention the margin of each product line as a potential driver of the profit problem(mix issue).
- Candidate should also infer about potential changes at the company in recent years and if these changes have impacted capacity, supplier relations, pricing, etc.
- A thoughtful framework, though unnecessary for this case's success, will include external pressures from competitors and potential changes in customer buying behaviors(peace time vs. war time, repeat customer purchases, fixed contract pricing, etc.)

Profit

Revenue
Price
Volume
Mix
Servicing Revenue? (if not clarified in prompt)

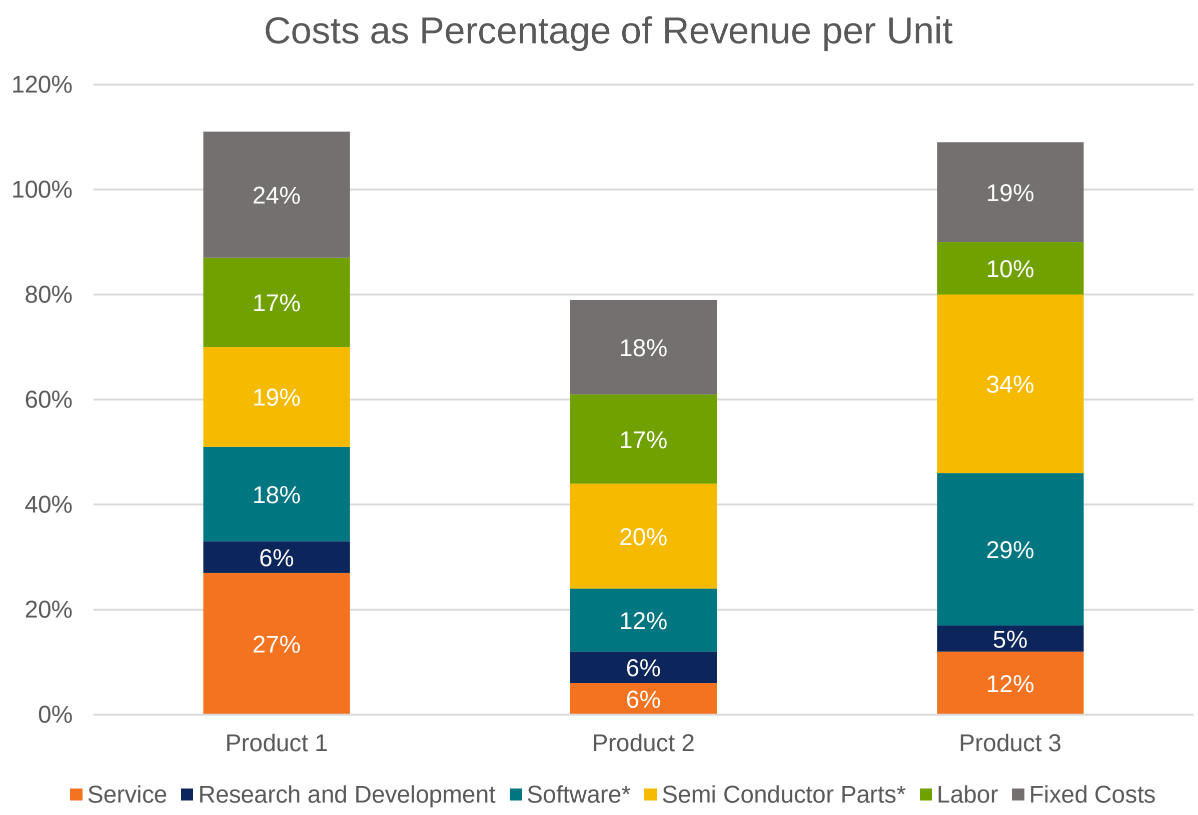
Cost
Labor
Materials
Software
R&D
Warranty
Overhead

Opportunities to explore
Remove a product line
Insource v. outsource
Value based price
Digitizing value chain
Expanding capabilities/product into other industries

Market
We serve one customer
War time v. peace time
Political influence of military budget
New competitors (deep tech)

- A powerful starting voice over of this framework states first I want to diagnose where the problem is occurring comparing our profitability YOY. Once we know, we can explore ideas for improving profitability. Of course, we also need to consider the unique market conditions for this company since we serve only one customer and they are the government. Then circling back to read through the details of each bucket.

EXHIBIT 1: REVENUES AND COSTS BY PRODUCT



Revenues and Historic Product Information	Product 1	Product 2	Product 3
Year of Product Launch	2017	2000	1986
Units Sold(2017)	5	10	5
Revenue per Unit(millions of $)	5	7	5
Cumulative Units Sold since Launch	5	185	520

* **Externally Sourced**

EXHIBIT 1 INSTRUCTIONS(FOR INTERVIEWER)

- Candidate should be able to drive where profitability/loss of company is coming from.
- **Question 1: Why do you think there are such significant differences in costs as a % of revenue?**
  - Product 1- new product, heavy on service early in product life cycle. High fixed costs relative to 2 and 3(opportunity to realize EOS with volume). Total costs are 111% of revenues.
  - Product 2- only profitable product. Highest revenue/unit. Model for what is working in the company. Total costs are 79% of revenues.
  - Product 3- old product. Heavily outsourced(few suppliers can supply the old tech needed, pricing bargaining power is not in Omega's favor). Total costs are 109% of revenues.
- **Question 2: Calculate profitability by product line and total current profitability for the firm.**

Profitability per unit(millions)	Product 1	Product 2	Product 3	Total
Revenue	5	7	5	
Total Cost	5.55	5.53	5.45	
Profit per unit	-0.55	1.47	-0.45	
Total Profit	-2.75	14.7	-2.25	9.7

- **Question 3: Conceptually, what would happen if you doubled the sales of product 1?**
  - Fixed costs would be shared across more units, cost per unit drops to 99% of revenue. All else equal, the product line becomes profitable.

BRAINSTORM AFTER EXHIBIT 1

- **Let's assume that Omega cannot sell more contracts. How would you determine areas for improvement in cost structure for each product line to raise profitability?**
  - Supply chain network assessment
  - Relationship pricing for externally sourced products
  - Compare to industry averages on labor
  - Identify historic cost behavior with other Omega product lines(most important inference).
    - When identified, provide exhibit 2.
    - If not mentioned, ask what you would look for in historic pricing to understand cost reduction opportunities?
      - Candidate should mention any patterns about cost behavior and savings in production over time from continuous improvement, economies of scale, software efficiency, etc.

EXHIBIT 2

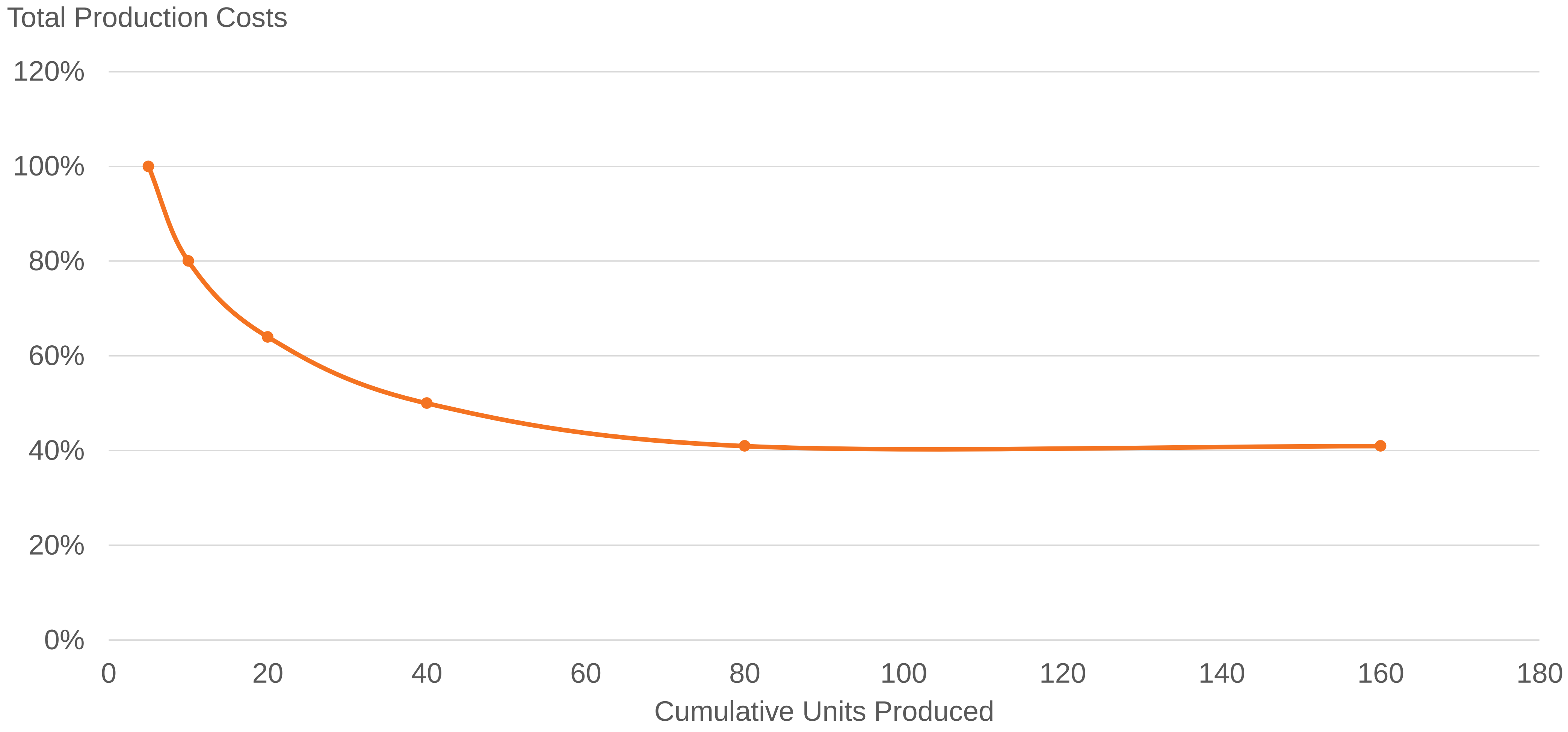

Total Production Costs
120%
100%
80%
60%
40%
20%
0%
0
20
40
60
80
100
120
140
160
180
Cumulative Units Produced

EXHIBIT 2(FOR INTERVIEWER)

- **Ask the candidate what he/she sees**
  - Every time a product doubles its cumulative production after its first 5 units, total production costs are reduced by 20%(help guide the candidate if the scale of the exhibit is not easy to establish)
  - Reduction in cost persists until 80 units are cumulatively produced and then no more improvement
  - (Referencing Exhibit 1) Product 1 has best opportunity for cost improvement. Accelerate production as quickly as possible to achieve profitability more quickly.
  - Product 2 has achieved its best cost behavior and peak profitability. Cash cow.
  - Product 3 is too expensive to continue producing. No more opportunity to reduce cost.
- **Knowing what we know, how should we proceed and why?**
  - Sunset production of Product 3 and absorb all sales to product 1. Accelerate cost reduction on Product 1
  - Risks of sunsetting Product 3?
    - Customer preferences, internal pushback, shutdown costs, production capacity for Product 1
  - **Now let's quantify the financial impact of this decision(next slide)**

CALCULATE FINANCIAL IMPACT(INTERVIEWER ONLY)

- Assume we can convert 100% of sales of Product 3 to Product 1 this year. Calculate the profitability improvement by product, in total, and net margin improvement as a percentage.
- Old Profitability

Profitability per unit(millions)	Product 1	Product 2	Product 3	Total
Revenue	5	7	5	
Total Cost	5.55	5.53	5.45	
Profit per unit	-0.55	1.47	-0.45	
Total Profit	-2.75	14.7	-2.25	9.7

- New Profitability(Product 1's new cost should now be 20% lower than at previous production, or 111% of revenue* 80%, or 88.8% of revenue)

Profitability per unit(millions)	Product 1	Product 2	Product 3	Total
Revenue	5	7	0	
Total Cost	4.44	5.53	0	
Profit per unit	.56	1.47	0	
Total Profit	5.6	14.7	0	20.3

- Change in Net Margin

	Old	New	Change
Total Profit	9.7	20.3	10.6
Net Margin	8.1%	16.9%	109%

RECOMMENDATION

- The candidate's recommendation should entail shutting down Product 3.
- Rationale should include improved EOS opportunities in Product 1, lack of profitability in Product 3, and opportunity for 100% conversion of customers due to the substitution of 1 for 3.
- Recommendation should quantify the improvement in $(10.6million) and %(109) terms.
- Risks should include monetary cost of shut down and customer pushback
- Next steps should include a cost analysis of the shutdown and/or communications plan internally and externally about the proposed plan

INTERVIEWER FEEDBACK FORM

Case Name ____________________________ **Interviewer** ______________________________

Case Book ______________________ **Case Type** _____________ **Difficulty** _____________

Case Execution:

- ❑ Clarifying Questions + Framework 1 2 3 4 5
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

Feedback:

- ❑ Exhibits + Quantitative Ability 1 2 3 4 5
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

Feedback:

- ❑ Brainstorm + Conclusion 1 2 3 4 5
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

Feedback:

- ❑ Presence & Non-Verbal 1 2 3 4 5
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

Feedback:

Total: ______ / 20

Boxing in the Opportunity

Private Equity

BOXING IN THE OPPORTUNITY

Private Equity

Prompt: Our client, GS Capital, is a private equity firm looking to acquire a new operating company, CorrugatedCo., which manufactures machines that cut, form, and print corrugated boxes. GS Capital has an opportunity to purchase the company for $100M and wants to know if it's a good idea.

Clarifying Information: *Note: Provide this only if corresponding questions are asked.*

1. Financial goal, metrics, timeline: GS Capital aims to achieve a 3.0-4.0x cash-on-cash return within five years. For simplicity, assume the transaction is completely funded by equity. (If candidate is still confused – In other words, GS Capital will pay for the company entirely with cash; it wants to sell the company in five years for at least 3-4 times more than it bought it.)

2. Geography: US only.

3. Business model: The company manufactures large, complex machines to sell to paper and packaging companies and then is exclusive provider of after market servicing and parts.

4. Understanding the product: Very large machine that can process 100 boxes per minute taking the box from raw corrugated board to final box you would see when you receive shipment from Amazon.

5. Other portfolio companies: We'll get into that later.

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a time you showed leadership on a team in a challenging situation.

Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions.

Current EBITDA

- Revenue
  - Machine (# of machines sold * average price)
  - Field Service (function of machine sales?, bill by hour
  - Part Sales (function of machine sales?)
- Costs
  - Raw Materials
  - Labor
  - Overhead
  - Warranty (big, complex machines)

Market

- Growth
  - GDP Growth
  - Disposable Income (consumer purchases could be a big driver of demand)
  - Industry Focus
  - International Expansion
- Competition
  - Fragmented
  - Barriers to entry
  - Quality differentiation

PE Firm

- Synergies
  - Other portfolio companies
  - Add-on acquisitions
  - Industry expertise
- Leadership Transition Plan
- Exit Strategy
  - Strategic
  - PE
  - IPO

How to Move Forward:

- Want to voice this over as first understand the current EBITDA scenario, then we can explore the market opportunities for growth, and finally want to consider we can probably bring value as well. Interviewee should drive toward finding current EBITDA first. If they move to financials then hand them exhibit 1.

EXHIBIT 1

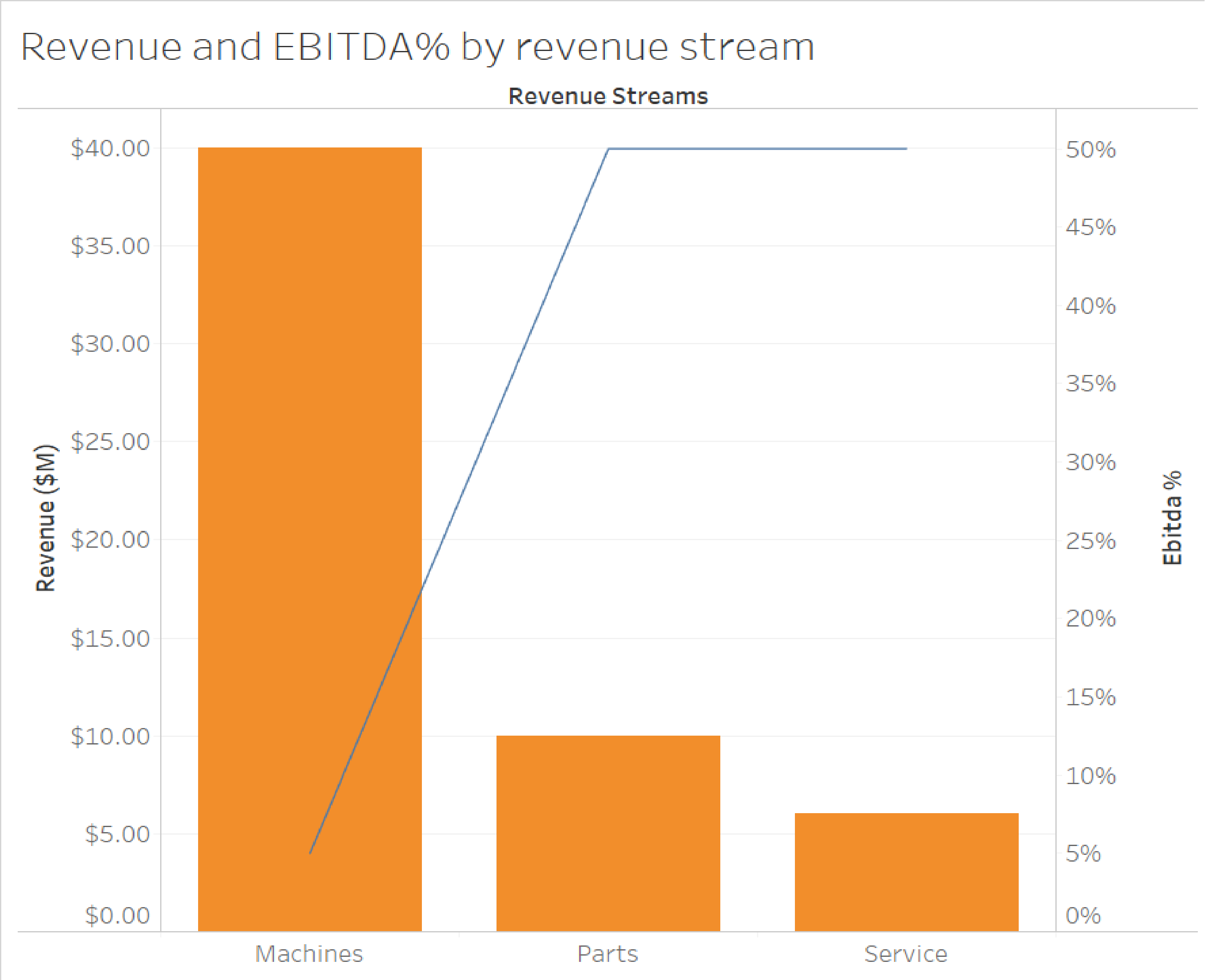
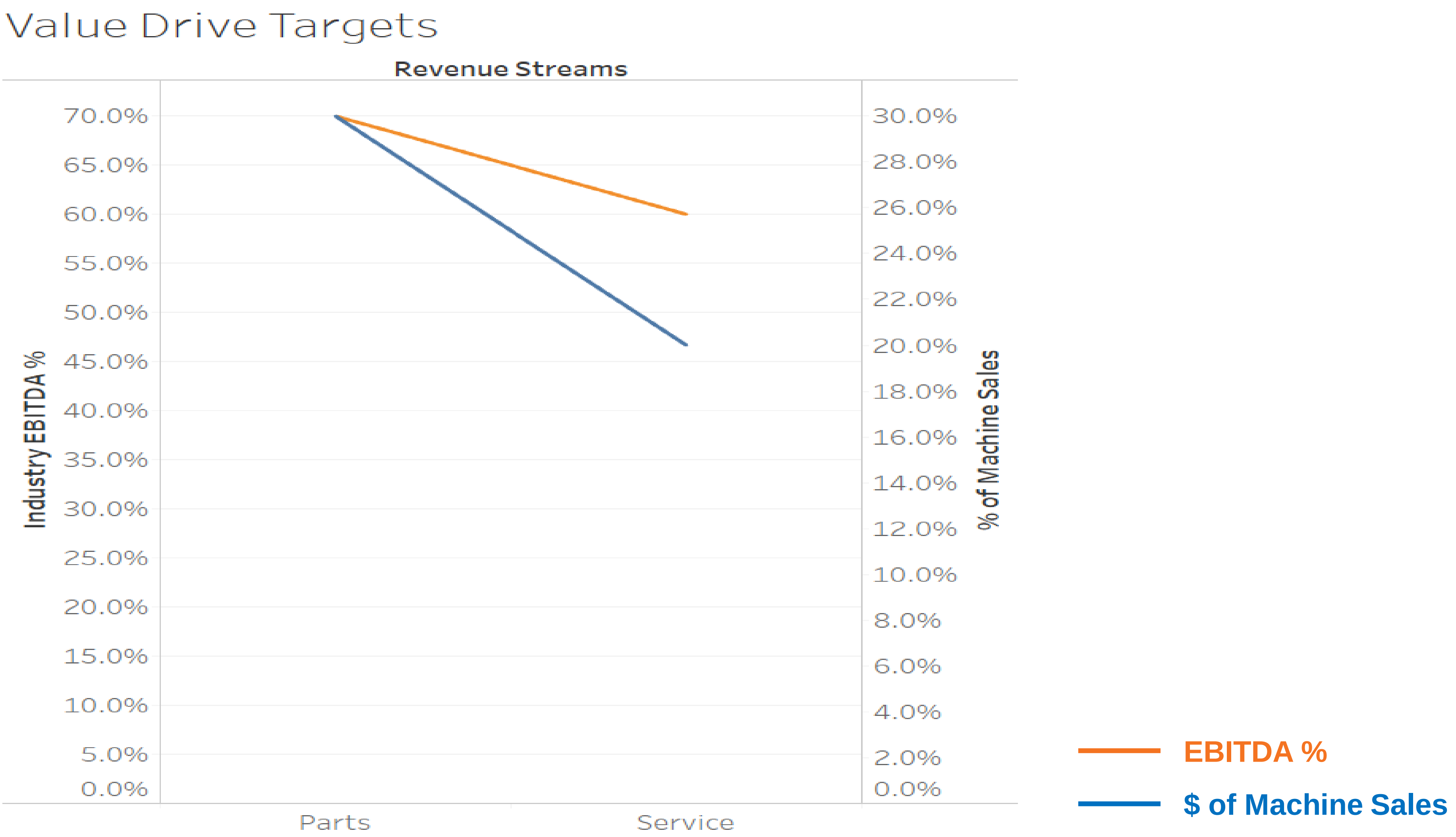

Revenue and EBITDA% by revenue stream
Revenue Streams
Revenue ($M)
$40.00
$35.00
$30.00
$25.00
$20.00
$15.00
$10.00
$5.00
$0.00
50%
45%
40%
35%
30%
25%
20%
15%
10%
5%
0%
Ebitda %
Machines
Parts
Service

EXHIBIT 2



*** Revenue based on % of average last three years of machine sales

EXHIBIT INSIGHTS

Exhibit 1

Machine revenue is largest revenues stream, but lowest EBITDA due to low EBITDA %

Candidate may be tempted to say focus on growing revenue of higher EBITDA% revenue streams, but key insight to this is understanding the business model that the machines build the customer base to drive revenues for the downstream revenue streams.

Drive forward should be understanding if your PE firm can drive growth in either EBITDA% or Revenue based on our PE capabilities, portfolio synergies, or purchasing add-on acquisitions

PE capabilities – Give them Exhibit 2

Portfolio Synergies – Tell them we have another company that does the folding, packing, and gluing of boxes. We believe machine sales market share will increase to 20% of the overall market by Y5 exit due to this portfolio synergy

Add-on acquisitions – no opportunities there

Exhibit 2

Two key pieces of information on this chart.

1. Industry EBITDA % averages for Parts and Field Service. Candidate should drive toward our ability to grow toward industry average and input that we are currently not there from Exhibit 1
2. Company can grow downstream sales as a % of Machine Sales Revenue. It is based on last 3 years of sales. Can assume we can get to that % by Y5.

MATH SOLUTIONS

	Market growth	5%					
	Market share	10%					20%
	Market size	400	420	441	463	486	510
		Y0	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**
Revenue	Machines	40					102
	Service	6					20
	Parts	10					30
EBITDA %	Machines	5%					5%
	Service	50%					60%
	Parts	50%					70%
EBITDA	Machines	2					5
	Service	3					12
	Parts	5					21
		10					**38**
						Multiple	10
Note: all numbers rounded to nearest whole number							380

- Candidate should recognize they need to get to Y5 EBITDA amount and then use a multiple.
- The multiple can be determined by imputing multiple from year 1 (10M EBITDA with $100M asking price). Should clarify with interviewer if okay to assume multiple does not change
- Candidate should ask about market growth and market share assumptions to get to year 5. Growth: 5%/year. Share: 10%->20%
- Candidate should realize they only need to calculate machine revenue for intermediate years since only Y5 EBITDA matters and all other numbers are function of machine revenue.
- Easiest way to get to Y5 Revenue is by dividing Y1 machine revenue from market share to get Y1 market share.
- Then growth market share by 5% per year to get to Y5 market share and then take 20% of that number for Y5 machine revenue.
- Service revenue = 20% of machine sales (per industry average)
- Parts revenue = 30% of machine sales (per industry average)
- Candidate should recognize that $380M meets target of $380M and then drive to thinking through risks of doing the investment.

BRAINSTORMING

What are the potential risks GS Capital may face with purchasing the company?

Brainstorming Guidance:

Note: This is just one possible set of categories and answers. Many more are possible, and interviewers should assess both the volume and relevance of answers.

Firm

- Leadership challenges (incumbent CEO disagreement and/or can't find go new leader
- Big, complex machines large warranty/liability exposure
- Cannot realize industry average EBITDAs
- Lose talented workforce when transition to PE firm
- R&D spending to keep advantage conflicts with cost efficient manner of PE firm

Market

- Major purchasers (Amazon switch suppliers) and they are not one of our customers
- Competitor emerges with superior technology
- International competitor emerges with lower labor costs
- PE Exit multiples decrease

CONCLUSION

Answer first…I recommend GS Capital purchases Corrugating Co. for $100M as we estimate they can sell it in 5 years for $380M, equating to a 3.8x cash-on-cash return. This falls on the high side of the funds target of a 3-4x return on cash.

Risks….I am concerned about the warranty exposure/expense for these machines…

Next steps… so would like to partner with you to create predictive algorithm for warranty exposure by machine.

INTERVIEWER FEEDBACK FORM

Case Name ____________________________ Interviewer ______________________________

Case Book ______________________ Case Type _____________ Difficulty _____________

Case Execution:

- ❑ Clarifying Questions + Framework — 1 2 3 4 5
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

Feedback:

- ❑ Exhibits + Quantitative Ability — 1 2 3 4 5
  - ❑ Accuracy
  - ❑ Speed
  - ❑ Insights Presented
  - ❑ Errors / Guidance Needed

Feedback:

- ❑ Brainstorm + Conclusion — 1 2 3 4 5
  - ❑ Creative & Structured
  - ❑ Good Business Judgment
  - ❑ Recommendation Strength

Feedback:

- ❑ Presence & Non-Verbal — 1 2 3 4 5
  - ❑ Confidence
  - ❑ Poise / Posture
  - ❑ Clear & Concise
  - ❑ Body Language
  - ❑ Coachability

Feedback:

Total: ______ / 20

To Automate or Not

Logistics

TO AUTOMATE OR NOT

Operations

Prompt:
After returning from a trade show, the CEO of a large grocery distribution center calls you. He enthusiastically describes a new technology which could be used to automate part of his company's process. He asks whether you think this would be a good idea for his business. Knowing that this CEO is a tech-enthusiast who loves innovation for the novelty of it, you ponder the implications. How would you tackle this problem?

Clarifying Information:

- The company does not have a specific goal in mind with this decision. This CEO trusts us and will do whatever we advise. This is to test the candidate's business judgment.
- Shipments are made to roughly 50 grocery stores in the immediate area, and the company does 1M shipments per year
- Costs to automate – (1) one-time outlay of $4M, plus (2) recurring OH, training, and additional maintenance costs of $1.0M (make the candidate request BOTH pieces of info)
- The candidate should visualize the distribution process (i.e. receiving, holding & picking, shipping) to think through this question

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a time where you made a mistake. What lessons did you learn?

2. What are three words that your learning team would use to describe you? Why?

Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions.

A strong framework could include:

Revenues

- Strong candidates will recognize that this is primarily a cost issue
- The company is not experiencing capacity issues and is satisfying customer demand, so optimizing the distribution process will not result in additional revenues

Costs

- Existing
  - Fixed Costs, Rents/Mortgage/Leases, Overhead, Salaries
  - Variable – Hourly employees, repairs& maintenance, inventory holding, shipping
- Incremental
  - One-time – Initial cost of automating, cost of initial training, downtime (there is none), severance pay for any employees terminated, etc.
  - Ongoing – Servicing, repairs, additional costs of running upgraded facility (i.e. energy, insurance, etc.)

Product

- Considerations around changes to product quality (there are none)
- Ability to distribute additional types of product, etc. (none)

Other

- Potential for future synergies (i.e. could the new technology interface with other future tech)
- Impact on relationships with employees, grocers, and/or suppliers

Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions.

A strong framework could include:

Guidance:

- If the candidate hasn't already visualized the process flow for a distribution center, press them to do so.
- After they've conceptualized the company's business tell them that the 3 phases of the company's distribution process are 1) Receiving, 2) Holding & Picking, 3)

Shipping Brainstorm:

- Move them into brainstorming exercise around the potential costs associated each phase:
- *Potential examples: Labor (time), equipment (forklifts, etc.), insurance, storage (i.e. cold and dry)*
- After the interviewer is satisfied with the depth and breadth of the brainstorming, provide Exhibit 1

EXHIBIT 1

Receiving

2,000 hrs.

Holding & Packing

10,000 hrs.

Shipping

3,000 hrs.

* All units are hours per week

Exhibit Guidance:

The interviewee should notice and verbally acknowledge the following in Exhibit 1. This is where you insert question and the answers to the question

- After confirming their understanding of the process flow, candidates should focus in on the total time required for the existing process
- Candidates should attempt to quantify the hours in terms of dollars of cost for each area of the process
- Info to provide upon request:
  - Each employee works 40 hours per week
  - Employee pay per hour is as follows: Receiving - $12/hr, Holding & Picking - $17/hr, and Shipping - $14/hr
- Once they obtain the above info, ask the candidate the calculate the total number of employees required in each function and the annual cost of labor (NOTE – Force them to use 52 weeks instead of 50 weeks)

	Total Hours	# Employees Required	Cost per Hour	Cost per Week
Receiving	2,000	50	$12.00	$24,000
Holding & Packing	10,000	250	$17.00	$170,000
Shipping	3,000	75	$14.00	$42,000
				$236,000
			x 52 weeks / yr.	
				$12,272,000

- After doing the calculations to the right, strong candidates will want to see how automating the process would change the data in exhibit 1
- Provide exhibit 2 when they ask for that info

EXHIBIT 2

Receiving

2,000 hrs.

Holding & Packing

5,250 hrs.

Shipping

6,000 hrs.

Math Guidance:

The interviewee should notice and verbally acknowledge the following in Exhibit 2. This is where you insert question and the answers to the question

- Once they obtain the above info, ask the candidate to calculate the change in the # of employees required in each function and the change in the annual cost of labor

	Total Hours	# Employees Required	Cost per Hour	Cost per Week
Receiving	2000	50	$12.00	$24,000
Holding & Packing	5250	131.5	$17.00	$89,250
Shipping	6000	150	$14.00	$84,000
				$197,250
			x52 per Yr.	
				$10,257,000

Current # Employees Required	Automated # Employees Required	Change
50	50	0
250	131.25	119
75	150	-75
	Employee Decrease	44

Current Costs: 12,272,000

New Costs: - 10,257,000

Annual Savings: 2,015,000

Math Guidance:

- Strong candidates should revert to the prompt and answer the initial question posed. This requires the candidate to compare the annual cost savings to the costs of automating.

Question 3:

- Ask the candidate to calculate a break-even in years for this investment:

Annual Savings	$2,015,000
Annual Incremental Costs	$1,000,000
Annual Incremental CM	$1,015,000
One Time Outlay	$4,000,000
Break Even Years	3.9

Ask the candidate whether a break-even of 5 years seems attractive

- There is no correct answer, and the Company does not have any policy on such matters. This should be used as an opportunity to test the candidate's business sense.
- Attentive candidates will consider the attractiveness of this payback period in light of the prompt. For example, if the CEO is potentially interested in the technology simply because of his love for innovation, would he really be willing to tie up his firm's cash for 5 years?

CONCLUSION

Recommendation:

- “The CEO is about to call you to ask for your recommendation. What would you say?”
- A strong recommendation will include:
  - Bottom Line – Automate or Do not automate
  - 2-3 Supporting Reasons
  - Risks pertaining to the bottom line
  - Next steps
  - Recommendations should be rooted in the prompt and should include numbers/data

INTERVIEWER FEEDBACK FORM

Case Name ______________________ **Interviewer** ______________________

Case Book __________________ **Case Type** ____________ **Difficulty** ____________

Case Execution:

- ❑ Clarifying Questions + Framework — 1 2 3 4 5
  - ❑ Good Questions
  - ❑ Structured
  - ❑ MECE
  - ❑ Creativity

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  - ❑ Body Language
  - ❑ Coachability

Feedback:

Total: _____ / 20

Rubber Bumper

Manufacturing

RUBBER BUMPER

Manufacturing

Rubber Bumper Co is a small family owned producer of rubber products. It prides itself on producing a limited range of products but producing the highest quality on the market. In general, new products are introduced after much deliberation and careful market study. The company has recently appointed a new President who noticed decreasing profits over the last couple of years.

(See the next page for Framework directions)

Clarifying Information:

Note: Provide this only if corresponding questions are asked.

What type of products do they sell? The company only sells two products; rubber bands and condoms

Is the company seeing similar declines in topline sales? Topline sales have remained relatively stable over the last 3 years

What is Rubber Bumper's market position? Rubber Bumper is the market leader in both of their product industries

BEHAVIORAL INTERVIEW QUESTION:

1. Tell me about a time you worked through an ambiguous problem.

2. Tell me about a time you had to convince someone of your idea.

Framework Guidance:

Note: There are many possible alternatives to this framework. These are only provided as possible suggestions.

Rubber Bumper Co has hired our firm to fix the decline in profits. What are all of the areas that need to be examined in order to identify any major issues that should be a priority?

How to Move Forward:

To get to the next portion of the case, the interviewee should ask to explore:

Expected Analysis: There are a number of possible frameworks for this question. A good answer will cover all areas you'd expect: **industry trends, margins, product mix, competitors, etc within reason. Generic frameworks are inappropriate.**

EXHIBIT 1

Rubber bands sold each year (millions of pounds of rubber)

	2011	2012	2013	2014	2015	2016	2017
Rubber Bumper	4	3	3	2.5	2.5	2.5	2
Max Rubber	17	19	21	21	22	23	24
Others (8)	9	9	8	7.5	6.5	4.5	5
Total	30	31	32	31	31	30	31

Condoms sold each year (millions of condoms)

	2011	2012	2013	2014	2015	2016	2017
Rubber Bumper	1	2	5	10	10	10	10
Spartan	100	110	108	115	117	115	115
Durable	150	155	152	158	159	165	170
Others (15)	99	93	105	107	119	130	155
Total	350	360	370	390	405	420	450

Exhibit Guidance:

The interviewee should notice and verbally acknowledge the following in Exhibit 1. This is where you insert question and the answers to the question

Question 1: The team decided to look at the product mix and their industry wide positions. The company only has two products: rubber bands and condoms. The analysts on our team compiled these two tables. (Display Ex. 1). What does this tell you?

How to Move Forward:

Expected Insights:

1. The rubber band market is flat whereas the condom market is showing strong growth in the United States
2. The dominant player in the rubber band industry is gaining more and more market share
3. While the condom industry is growing (30% from 2005 to 2011) the major competitors are not growing as fast (~15% each).
4. The condom industry is more fragmented than the rubber band industry, and the smaller players are getting a larger proportion of the market

Commentary:

The candidate should ultimately start to see that the rubber band industry is becoming less attractive and the condom industry is showing growth and the major market players are not keeping up with the growth. A great response will automatically want to see why Rubber Bumper’s condom growth has tapered off while the industry keeps expanding. If they do not get to that issue, prod them until they do.

Question 2: Rubber Bumper Co has two factories, each producing one of their two products. They essentially do not share any fixed costs and for the most part are run as separate P&L's. The team would like you to put together a quick summary and compare the financial profitability of each of the two factories for the most recent year.

Rubber Band Factory

- They make boxes of 500 rubber bands that they sell to retailers for $20 a box
- 1 pound of rubber makes approximately 125 rubber bands
- They should already have the amount of rubber they used from the exhibit
- The rubber band factory has an inclusive $4MM in annual overhead
- **[ONLY PROVIDE WHEN ASKED]** It costs $1 to turn a pound of rubber into a pound of rubber bands (assume no waste)

Condom Factory

- They sell 4 packs of condoms to retailers for $3 a pack
- They factory is smaller than the rubber band factory and only costs $2 MM in annual overhead, inclusive of everything
- **[ONLY PROVIDE WHEN ASKED]** Each condom costs $0.10 to make

Expected Calculations: (Question 2)

Students may get tripped up because one starts with the number of condoms, and the other starts with the amount of pure rubber and need to infer the number of bands sold. (rb = rubber band):

- 2MM lbs of rubber x 125rb's/lb = 250 MM rb's
- 250MM rb's / 500 rbs/box = 500K boxes x $20 box = $10MM in Revenue
- 2MM lbs of rubber x $1 = $2MM in variable costs
- RB Profit = $10 - $2 - $4 = $4MM in profit
- 10 MM condoms / 4 pack = 2.5 MM packs x $3 = $7.5MM in Revenue
- 10MM condoms x $0.10 = $1MM in variable costs
- Condom Profit = 7.5 – 1 – 2 = 4.5MM in profit

Commentary: A good answer will arrive at the math and note that the two profits are fairly comparable. A great answer will also note that the margin on the condoms is significantly higher (4.5 MM in profit for 3 MM in costs, rather than 4MM in profit for 6 MM in costs). Also, a great answer will question why the overhead for the condom factory is small compared with the rubber band.

Question 3: It turns out that the two overhead figures are significantly different because the capacity of the second factory is much smaller, about half as much. The President has asked us to evaluate whether we should switch production of the first plant to make more condoms since we have reached capacity at the smaller plant (assume a factory can only make 1 thing).

Supplementary Information

Give out the following information ONLY WHEN REQUESTED.

- It will cost $2MM dollars to refurbish the new plant and take 1 year to complete during which time the factory will be off line. (If asked, assume there are no tax benefits from depreciating the CapEx and no cost of capital)
- Overhead would remain the same
- During this time, we won’t be able to make any rubber bands
- The bigger plant can produce twice the volume of condoms as the smaller plant.
- Rubber Bumper Co’s payback period for such projects is 4 years.
- Assume that Rubber Bumpers rubber band demand has stabilized at 2MM lbs per year.

Expected Calculations: (Question 3)

A bad answer will simply look at 4.5MM in profit and 4.0MM and say that Rubber Bumper should make more condoms. A great answer will look into the costs, the opportunity costs, and the payback period to evaluate whether this should go forward. Also a great interviewee will recognize the timing of the payments, and while we don't have a cost of capital there is a timeline such investments must prove profitable.

- 1 year offline they are losing (10MM – 2MM) = $8MM in contribution
- Capital Expenditures = $2MM
- Total Cost of Project = 8 + 2 = $10MM

Note we assume that overhead will be paid while the factory is offline, but it should not count as an additional cost, since we would pay that anyways.

- The benefit is the difference in profitability between the two products.
- The bigger factory can produce twice as many condoms; 6.5MM x 2 = 13MM
- The bigger factory is currently producing 8MM in contribution (because we are looking for the difference, the $4MM of overhead is a wash)
- Switching will create an additional $13 – 8 = 5MM in profit.
- The interviewee should draw a conclusion towards the end noting that under these assumptions the project will repay itself in year 3 (1 year offline + 2 years of operation) and that it is within the required time frame. Additional second level insights are encouraged.

BRAINSTORM

What are some of the risks involved with this project?

Brainstorming Guidance

This is a “what else” section. Below are some basics but ideally you’re looking for the interviewee to be as creative as possible. As with most questions of this type, a bad answer will stop at one or two. A good answer will have a creative list. A great answer will have a structure that makes the answer MECE. A great answer should also prioritize the findings indicating which ones he thinks are the most important.

Key Points to Consider

- Assumes that Rubber Bumper can sell 3x the number of condoms it sells today, immediately.
- Assumes that rubber band demand wont rebound. The bigger plant is equally profitable because it is being underutilized
- Political parties could kill sex ed.
- Less diversification in products exposes them to increased market risk
- Condoms are not as generic of a product as rubber bands and may require a larger investment in advertising to compete on a higher level
- Potentially more legal risk in selling contraception than rubber bands
- Employees may not want to make condoms.

CONCLUSION

The President is walking in the board room and expects a summary. Please summarize your findings.

Expected Analysis

The summary should start with a recommendation. “You should convert the plant” and then back track into the reasoning: industry trends and financial justification. Finally it should mention which of the risks are the most problematic and how he would mitigate it. The interviewer should feel free to challenge any part of the conclusion and expect a well worded response.

INTERVIEWER FEEDBACK FORM

Case Name ______________________ Interviewer ________________________

Case Book __________________ Case Type ___________ Difficulty ___________

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